

Optimal-stopping recourse for vehicle routing with simultaneous pickup, delivery, and stochastic demands

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Abstract

In simultaneous pickup-and-delivery operations the load of a vehicle evolves stochastically along its route, and a mid-route capacity breach forces costly emergency recourse: ~~a surge-priced replacement vehicle and compensation for late customers~~. Fleets hold two cheaper levers—handing the remaining route to a standby vehicle, or detouring to the depot to unload collected pickups—whose value depends on the vehicle’s position and on prices built from the route’s real geometry. We cast the ~~question-choice~~ of when to exercise which lever as a finite-horizon optimal-stopping problem ~~under an empirical demand distribution, solved by and solve it with~~ BATON, ~~a policy that which adapts regression-based backward induction to this setting: it estimates per-stop continuation costs by backward induction over historical scenarios and intervenes exactly from historical demand scenarios with a monotone regression, a shape we prove the true continuation value satisfies, and intervenes~~ when continuing becomes dearer than the cheapest recourse action. ~~The policy has no tunable threshold; we prove that no fixed threshold reproduces the optimal intervention boundary under position-dependent prices~~ We show that endpoint-based risk labels are blind to mid-route peaks, that

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every fixed threshold over-triggers relative to the optimal rule, and that the endpoint-based risk labels of prior work are blind to mid-route peaks. Across price of this over-triggering is bounded by the probability of interventions on days that would have completed cleanly. On the Dethloff and Salhi–Nagy benchmarks and new instances on new instances built on the road networks and real retail locations of five cities, BATON dominates published recourse policies—rule-based restocking triggers has the lowest expected cost of the implementable policies compared—rule-based and position-dependent thresholds, depot-return restocking, rollout, and a reinforcement learning policy—under reinforcement learning—under six planning regimes including three published robust planners. It attains 92–99% of, and reaches 86%–96% of the saving of a near-exact dynamic program given fifty times its data, exceeds the handoff-only clairvoyant bound by pricing the full. Because the depot return enlarges the action set, adapts its recourse mix to fleet economies without re-tuning, and runs it can exceed the clairvoyant bound of the handoff-only class; on real urban networks the return rarely pays and the policy reduces to its handoff-only form. The conclusions survive dependent demand, non-exchangeable days and a capacitated standby pool shared across routes. The policy needs no tuning and fits in milliseconds per route; all underlying routing plans are certified by mixed-integer programming bounds.

Keywords: Routing, stochastic vehicle routing, optimal stopping, recourse policies, last-mile logistics

1. Introduction

Every morning, an urban parcel operator loads a fleet of vehicles with the day’s deliveries and sends each one along a planned sequence of stops. In the simultaneous pickup-and-delivery mode that dominates e-commerce logistics in dense Asian cities—and increasingly elsewhere—each stop is both a delivery and a collection: the driver hands over parcels and takes returns,

exchanges, or outbound shipments on board. This operational convenience has an awkward physical consequence. The load of the vehicle no longer decreases monotonically along the route, as it would in a pure delivery tour; it drifts downward with deliveries and jumps upward with pickups, and on an unlucky day the *running peak* of this load exceeds the vehicle’s capacity somewhere in the middle of the tour, even though the final load—everything collected minus everything delivered—would have fit comfortably. When that happens, the operational damage is immediate and expensive. The vehicle can serve no further stop until it is relieved; an ad-hoc replacement must be hired at surge prices to complete the route; and every remaining customer is served late, incurring service-level compensation and goodwill loss. Operators therefore watch their vehicles’ loads in real time and, when a breach looks imminent, exercise one of two cheaper forms of recourse. They can *hand off* the remainder of the route to a standby vehicle from a pooled reserve, paying a callout, a per-kilometre surcharge on the remaining tour, and a cargo-transfer dwell. Or, when the depot is near, they can order a *depot return*: the vehicle swings by the depot, unloads the pickups collected so far, reloads its remaining deliveries, and continues its own route, delaying its downstream customers by the detour.

The decision problem this creates is deceptively simple to state and surprisingly subtle to solve. After every stop, given what the vehicle has observed of today’s demands so far, should it continue, hand off, or return to the depot? Acting too early wastes money on recourse that hindsight shows was unnecessary—and standby capacity is not free, since every handoff consumes a vehicle-day from the reserve pool. Acting too late means paying the full price of a breach. The trade-off shifts along the route, because both recourse prices fall as fewer stops remain while the scope for demand surprises shrinks; it shifts with the geometry, because a depot return is nearly free downtown and prohibitive at the urban fringe; and it shifts with the day, because the demands observed at the first stops carry information about the demands

still to come. A policy that captures all three effects must, at every stop, weigh the *cost of continuing optimally*—an expectation over everything that might still happen, valued under the best future use of both levers—against the current prices of acting now.

The recourse literature for stochastic vehicle routing has approached this class of decisions almost exclusively through *threshold rules*. In the vehicle routing problem with stochastic demands (VRPSD), the classical preventive-restocking policies return the vehicle to the depot when its residual capacity falls below a threshold—a fixed fraction of capacity, a multiple of the next customer’s expected demand, or a multiple of the expected demand remaining on the route (Yang et al., 2000; Salavati-Khoshghalb et al., 2019). Threshold rules are attractive because they are transparent and easy to implement, and with carefully tuned parameters they capture a useful share of the available saving. But they hard-wire two approximations that our setting exposes. First, a threshold on the current load ignores *option value*: early in the route, waiting is nearly free because many opportunities to act remain, while late in the route the same load level may leave a single chance to intervene. A fixed threshold cannot be right at both ends, and we prove below that its error is one-sided—it always over-triggers. Second, and more fundamentally, when recourse prices vary by stop—as they do whenever they are built from real route geometry—the optimal intervention boundary is not representable by *any* single threshold on load or on breach probability: the comparison that defines optimal behaviour is between two costs, not between a probability and a constant.

This paper develops, analyses, and ~~extensively~~ benchmarks a policy that makes exactly that comparison. BATON (*Backward-induction AcTion pricing for ONline recourse*) learns, from nothing more than historical demand realisations, a per-stop estimate $\hat{C}_k(w)$ of the expected cost of continuing past stop k with load state w under optimal future play, by sweeping the historical paths backwards in the manner of Longstaff and Schwartz (2001) and

projecting each regression onto the class of monotone functions—a shape restriction we prove the true continuation value satisfies when the load evolves as a stochastically monotone Markov chain. Online, the vehicle performs a single comparison after each stop: it exercises the cheapest recourse action whose price is below the estimated cost of continuing, and otherwise drives on. The policy has no tuning parameter ~~at all~~; the threshold rules of the literature are recovered as its myopic degeneration. ~~Crucially, the~~ The same backward induction prices ~~an arbitrary menu~~ a menu of recourse actions. ~~Adding:~~ adding the depot return requires one extra term in the Bellman recursion—the continuation value from the post-return state, which the algorithm evaluates ~~exactly~~ by simulating the reset state through its own already-fitted downstream policy—and the resulting three-action policy chooses, stop by stop and day by day, whichever lever the economics favour.

~~We subject this policy to what we believe is the most comprehensive computational study of~~ The paper makes three contributions, and we separate them because they are of different kinds. The first is a *model*: execution-stage recourse in ~~the~~ simultaneous pickup-and-delivery ~~setting to date~~. ~~The study rests~~ routing, with its non-monotone load, formulated as a finite-horizon optimal-stopping problem over a menu of levers whose prices are built per stop from the route’s geometry and a three-class fleet economy. The second is a set of *structural results* about that model: endpoint-based risk labels are blind to mid-route peaks, and we bound the excess cost this causes (Proposition 1); the stopping region of any myopic or fixed-threshold rule contains the optimal one (Proposition 2), and the price of this over-triggering is bounded by the probability of interventions on days that would have completed cleanly (Proposition 3); and the continuation value is monotone in the load (Proposition 4), which justifies a nonparametric monotone estimator. The third is a *method* that is deliberately an adaptation rather than an invention: regression-based backward induction for optimal stopping (Longstaff and Schwartz, 2001) is established, and we claim novelty only for the pieces the model requires—the

isotonic regression step licensed by Proposition 4, the simulation-based valuation of the post-return state, and the train-set selection of the recourse menu.

We evaluate the policy in an extensive computational study resting on three pillars. The first is breadth of comparison: we evaluate against the published rule-based restocking family of [Salavati-Khoshghalb et al. \(2019\)](#), a depot-return restocking policy in the tradition of [Florio et al. \(2020\)](#) and [Legault et al. \(2025\)](#), a rollout policy in the spirit of [Secomandi \(2001\)](#) and a cost-scaled version of it, a tuned threshold ~~given on~~ our corrected risk labels and a position-dependent threshold with one tuned level per stop, a reinforcement-learning policy re-implemented from [Iklassov et al. \(2024\)](#) ~~and trained on a GPU, and two~~, and dynamic-programming benchmarks—one restricted to the same data budget as BATON and ~~one two~~ supplied with fifty times more data as ~~a near-exact yardstick~~ yardsticks. The second pillar is breadth of environment: six planning regimes generate the routes, from a deterministic gate through sample-average and Wasserstein distributionally robust gates to three published robust planners ([Gounaris et al., 2013](#); [Bertsimas and Sim, 2004](#); [Ghosal et al., 2024](#)); the instances span the classical Dethloff and Salhi–Nagy benchmarks and a new set built on the actual road networks of Ho Chi Minh City, Hanoi, New York, Paris, and Shanghai, with customers placed at real shop locations ~~so that the spatial demand distribution is the true retail one~~. The third pillar is anchoring: the execution stage is compared against ~~exact~~ near-exact dynamic programming, the planning stage is certified against mixed-integer programming bounds computed with a commercial solver, and a clairvoyant bound measures how much saving is achievable at all.

~~The empirical picture that emerges is unambiguous.~~ The study also probes the assumptions behind the model—demand dependence, non-exchangeable days, the post-return state, a capacitated standby pool shared across routes, and the size of the training history.

BATON attains the lowest expected execution cost of every implementable

policy on every planning gate and every benchmark ~~family, with paired significance levels that leave no room for doubt; on~~ family—the only exception being a tie within a few tenths of a point on a deliver-only city variant where recourse is worth least—and reaches 86%–96% of the saving of the near-exact three-action dynamic program that sees fifty times its data. ~~On~~ four of the six gates it exceeds the clairvoyant bound of the handoff-only class; ~~which is the cleanest possible demonstration that pricing a richer action set, not statistical luck, is the source of the gain.~~ This is not a paradox: that bound is clairvoyant only about the handoff lever, and a depot return can prevent a breach at a price no handoff can match, so the excess measures the value of the second lever. The economics sweeps deliver the managerial headline: when standby capacity is expensive, threshold policies destroy value while BATON ~~quietly~~ shifts its recourse mix toward depot returns; when service-level penalties rise, it shifts back—all without re-tuning, because the policy prices actions rather than memorising trigger levels. ~~Two deliberate negative results round off the study: enriching the conditioning statistic beyond the scalar load state does not help at operational data scales, and quadrupling the training budget of the reinforcement-learning baseline does not close its gap.~~ The gains are not uniform, and we report where they are small: on real urban networks the depot return rarely pays and BATON reduces to its handoff-only form, and on tightly packed routes a well-tuned threshold comes close to the handoff-only policy, as Proposition 3 predicts.

The remainder of the paper unfolds as follows. Section 2 positions the work within the recourse, robust-planning, and learning literatures. Section 3 develops the operational setting and cost model, formulates the stopping problem, diagnoses why endpoint labels and fixed thresholds fail, and presents the BATON policy with its supporting analysis and algorithms. Section 4 reports the computational study. Section 5 distils the managerial insights, ~~acknowledges~~ states the scope of the conclusions and their limitations, and concludes.

2. Related work

The vehicle routing problem with stochastic demands is one of the oldest and most studied stochastic combinatorial problems in operations research; the surveys of [Gendreau et al. \(1996\)](#) and [Oyola et al. \(2018\)](#) trace its development, and [Koç et al. \(2020\)](#) review the simultaneous pickup-and-delivery variant that concerns us. Within this literature our work sits at the intersection of three strands: recourse policies for the execution stage, robust and distributionally robust planning for the construction stage, and learning-based approaches that estimate policies from data.

The recourse strand begins with the a priori optimisation paradigm of [Bertsimas \(1992\)](#) and [Dror et al. \(1989\)](#), in which routes are fixed before demands are revealed and a simple rule—return to the depot upon failure—prices the risk. The insight that *preventive* action dominates reactive failure recovery led to the restocking policies of [Yang et al. \(2000\)](#), who compute optimal return thresholds by dynamic programming on a fixed route, and to a rich family of rule-based triggers whose modern form is analysed by [Salavati-Khoshghalb et al. \(2019\)](#): return when residual capacity falls below a fixed fraction of capacity, below a multiple of the next customer’s expected demand, or below a multiple of the expected demand remaining. Exact algorithms that price optimal restocking within route optimisation culminate in [Florio et al. \(2020\)](#) and, most recently, in superadditivity-based formulations ([Legault et al., 2025](#)) and scenario-optimal recourse under empirical demand distributions ([Ota and Fukasawa, 2026](#))—the latter sharing our data-driven stance, though at the planning rather than the execution stage. Reoptimisation approaches solve the execution stage as a Markov decision process: [Secomandi and Margot \(2009\)](#) characterise partial-reoptimisation policies, and the rollout algorithm of [Secomandi \(2001\)](#) remains the standard implementable approximation, improving a base policy by one step of policy iteration. Our contribution to this strand is a policy that retains the implementability of the rule-based family while provably escaping its two

structural approximations—the endpoint-for-peak label substitution and the threshold functional form—and that generalises across recourse menus, pricing the depot return of the restocking tradition and the vehicle handoff of platform logistics within one recursion.

The robust-planning strand constructs routes that remain feasible when demands deviate. [Gounaris et al. \(2013\)](#) formulate the robust capacitated VRP under polyhedral demand uncertainty and derive the quadrant-budget sets we use as one of our planning gates; [Gounaris et al. \(2016\)](#) give an adaptive memory metaheuristic, and [Pessoa et al. \(2021\)](#) the state-of-the-art exact algorithm, for the same problem class. Budget uncertainty in the sense of [Bertsimas and Sim \(2004\)](#) supplies a second gate, and the unifying distributionally robust framework of [Ghosal et al. \(2024\)](#), whose moment-ambiguity feasibility check reduces to a Cantelli-type closed form, a third. Sample-average and Wasserstein-ambiguity gates ([Mohajerin Esfahani and Kuhn, 2018](#)) complete the planning spectrum. We treat these planners as *environments* rather than competitors: execution recourse operates on whatever routes planning produces, and one of our findings is that planning conservatism and execution intelligence are substitutes at the margin—conservative plans leave little for any execution policy to save, yet it is precisely there that pricing the full action menu matters most, because on such plans the threshold family destroys value while the priced policy still extracts double-digit savings.

The learning strand estimates routing policies from data. Approximate-dynamic-programming approaches with lookup tables ([Novoa and Storer, 2009](#)) and offline value-function approximation ([Ulmer et al., 2019](#)) established that learned continuation values can guide online routing decisions; the recent wave of neural constructive methods extends this to end-to-end route construction, with [Iklassov et al. \(2024\)](#) the representative treatment of the stochastic demand case and [Kadyrov et al. \(2025\)](#) a recent application to dynamic settings; [Hu and Tao \(2025\)](#) and [Wang et al. \(2026\)](#)

study uncertainty in simultaneous pickup-and-delivery variants adjacent to ours. Our method is deliberately conservative within this strand: it uses the regression-based backward induction of American-option pricing (Longstaff and Schwartz, 2001; Tsitsiklis and Van Roy, 2001), whose convergence is understood (Clément et al., 2002), combined with isotonic regression (Barlow et al., 1972; Ayer et al., 1955) whose shape constraint we justify structurally. Two of our empirical results speak directly to this strand: a policy-gradient network trained on ~~a GPU the same data~~ captures materially less of the achievable saving than the monotone regression ~~from identical data~~, and enriching the conditioning statistic with factor posteriors and recency features—the natural representation-learning move—makes matters slightly worse at operational sample sizes. ~~Simple, shaped, and exhaustively priced beats flexible and learned in this regime, and we consider establishing that boundary to be part of the paper’s contribution~~ We read these results as statements about a regime, not about learning in general: with a scalar sufficient state, a known monotone value structure and a thousand paths per route, a shaped regression is hard to beat, and §4.4 discusses which learning architectures could narrow the gap as the state or the data grow.

3. The execution problem and the BATON policy

This section develops the paper’s methodological content as one continuous argument: the operational setting and its cost structure (§3.1), the optimal-stopping formulation (§3.2), the two structural defects of the incumbent approach and their formal diagnosis (§3.3), the estimation scheme (§3.4), the generalisation to the full recourse action set (§3.5), and the resulting algorithms with their complexity and reference bounds (§3.6). The four propositions have been verified formally in the Lean 4 proof assistant with the Mathlib library; the formal statements, and how they correspond to the text, are part of the replication repository (see Data availability). The formalisation of the inclusion and strictness claims of Proposition 2

and of Proposition 3 works on finite probability trees whose nodes are full histories. On finite-support demand distributions—including the empirical distribution of the training days—these claims therefore hold without the Markov Assumption 1, with continuation values conditioned on the history rather than on W_k alone.

3.1. Operational setting and the three-class fleet cost model

Consider one planned route serving m customers in a fixed sequence $1, \dots, m$ from a depot. The vehicle departs carrying the route’s deliveries; at stop k it delivers a random volume $d_k \geq 0$ and collects a random pickup $p_k \geq 0$, each observed on service, producing the *net increment*

$$g_k = p_k - d_k. \quad (1)$$

The *running net load* is

$$W_k = \sum_{j=1}^k g_j, \quad W_0 = 0, \quad (2)$$

and with departure load L_0 and capacity Q the vehicle physically overflows at stop k when $L_0 + W_k > Q$, that is, when W_k exceeds the *slack* $B = Q - L_0$. The *breach time* is

$$T = \min\{k \leq m : W_k > B\}, \quad \min \emptyset := \infty, \quad (3)$$

and the route *peak-overflows* on the event $\{T \leq m\} = \{\max_{k \leq m} W_k > B\}$. We write $\mathcal{F}_k = \sigma(g_1, \dots, g_k)$ for the information available after serving stop k .

Costs follow the economics of an urban last-mile operator running three vehicle classes. The *planned* fleet is contracted at a day rate F_{plan} per vehicle plus a running cost c_{km} per kilometre; these are sunk once the plan is fixed and play no role in execution decisions. The *standby* class supplies

planned handoffs: capacity is pooled across the operation, and a handoff after stop k consumes one standby vehicle-day at rate F_{sb} , a dispatch fee, a per-kilometre surcharge s_{ho} on the remaining tour δ_k (the distance of legs $k \rightarrow k+1 \rightarrow \dots \rightarrow m \rightarrow \text{depot}$), and a cargo-transfer dwell c_{tr} :

$$H_k = F_{\text{sb}} + F_{\text{ho}} + s_{\text{ho}} c_{\text{km}} \delta_k + c_{\text{tr}}. \quad (4)$$

The *emergency* class is hired ad hoc at surge prices when a breach actually occurs at stop k : a surge callout F_{emg} , the remaining distance at a surge multiplier s_{emg} , a service-level compensation p_{late} for each of the $m - k$ downstream customers served late, and a goodwill loss at the breached stop itself,

$$E_k = F_{\text{emg}} + s_{\text{emg}} c_{\text{km}} \delta_k + p_{\text{late}} (m - k) + p_{\text{breach}}. \quad (5)$$

Finally, a *depot return* after stop k sends the vehicle from stop k to the depot and back to stop $k+1$, emptying the pickups collected so far and reloading the plan’s remaining deliveries; it costs the detour at the running rate plus the transfer dwell plus the same per-customer lateness as any other delay,

$$R_k = c_{\text{km}} (D_{k,0} + D_{0,k+1} - D_{k,k+1}) + c_{\text{tr}} + p_{\text{late}} (m - k), \quad (6)$$

where $D_{i,j}$ denotes road distance. All three schedules are computed from the route’s actual geometry before departure and are therefore known constants at decision time; both H_k and E_k decrease along the route, ~~and~~. At our default parameters

$$E_k > H_k \quad \text{for all } k, \quad (7)$$

~~since~~ an unplanned breach at a stop ~~is always being~~ dearer than a planned transfer at the same stop. Inequality (7) describes the calibration, not a requirement of the method: none of our results uses it (Assumption 2 below asks only that E_k be non-increasing), and where it fails—when standby capacity is priced above an emergency hire, as in one configuration of §4.8—the

handoff is simply dominated at the stops concerned and the policy never uses it there. The flat two-price model $H_k \equiv \omega_F$, $E_k \equiv C_{\text{fail}}$ with $C_{\text{fail}} > \omega_F$ is the special case used in our controlled experiments and by the incumbent system.

Standby versus emergency vehicles. The two recourse classes differ in how they are contracted and in what they cost the customer, not in the vehicles themselves. A *standby* vehicle is reserved capacity under a pre-agreed rate card: it is dispatched *before* a breach, meets the planned vehicle at the next stop, takes over the remaining route at a small per-kilometre surcharge, and keeps the downstream customers on schedule apart from the transfer dwell. An *emergency* vehicle is hired on the spot market *after* the planned vehicle has overflowed: it is paid a surge callout and surge distance rate, must first reach a vehicle that is already stranded, and every downstream customer is served late. The standby day rate F_{sb} in (4) is the holding cost of standby capacity, charged per vehicle-day actually consumed. This pay-per-use convention treats the standby pool as shared across a metropolitan operation and large relative to one plan, so that a single plan’s handoffs never exhaust it; it is the standard arrangement with third-party reserve fleets, and it makes the pool size a by-product of the policy (the 95th percentile of daily handoffs, reported with our results) rather than a constraint on it. When instead a fixed number of standby vehicles is reserved in advance, their holding cost is sunk at execution time and the pool is a shared, capacitated resource that couples the routes; §4.6 treats that case explicitly.

3.2. The optimal-stopping formulation

Restrict attention for the moment to the handoff lever alone; the general action set is treated in §3.5. An *execution policy* is then a stopping time σ with respect to $(\mathcal{F}_k)$ taking values in $\{1, \dots, m - 1\} \cup \{\infty\}$, with $\sigma = \infty$

meaning never hand off, and the realised execution cost of a day is

$$J(\sigma) = \begin{cases} E_T, & T \leq \sigma \wedge m \quad (\text{breach first}), \\ H_\sigma, & \sigma < T \quad (\text{handoff first}), \\ 0, & \sigma = \infty, T = \infty \quad (\text{clean completion}). \end{cases} \quad (8)$$

The operator seeks $\inf_\sigma \mathbb{E} J(\sigma)$, the expectation taken over the demand law. Throughout the analysis we assume:

Assumption 1. ~~The increments $g_1, \dots, g_m$ are independent (not necessarily identically distributed), so that (W_k) is Markov and W_k is a sufficient statistic for the future of the load path~~ running load $(W_k)_{k=0}^m$ is a Markov chain with respect to $(\mathcal{F}_k)$, possibly time-inhomogeneous, whose transition kernels are stochastically monotone: for every k and all $x \leq x'$, the conditional law of W_{k+1} given $W_k = x$ is dominated in the usual stochastic order by its conditional law given $W_k = x'$.

Assumption 2. ~~$H_k \leq E_k$ for all k , and both schedules are~~ The emergency schedule (E_k) is non-negative and non-increasing in k ; the handoff and return schedules $(H_k), (R_k)$ are arbitrary non-negative constants.

Assumption 1 ~~matches the benchmark protocol exactly when demands are independent and holds approximately under correlated demands, where the running sum absorbs a common demand factor; the correlated case is examined empirically in §4~~ makes (k, W_k) a sufficient state and covers two cases of practical interest. The first is independent increments, not necessarily identically distributed: the conditional law of W_{k+1} given $W_k = x$ is that of $x + g_{k+1}$, which is stochastically increasing in x . The second is positively dependent demand driven by a common day-level factor: if $g_j = \mu_j + \beta Z + \varepsilon_j$ with a Gaussian factor Z and independent Gaussian noise ε_j of common variance, the likelihood of $(g_1, \dots, g_k)$ depends on Z only through W_k , so W_k is sufficient for the factor, the posterior of Z is increasing

in W_k , and the chain is Markov with stochastically monotone kernels. Outside these cases W_k is not exactly sufficient—in our benchmark protocol pickups and deliveries load on separate factors with customer-specific scales—and the policies of this paper are then optimal within the class of policies that observe (k, W_k) only in the approximate sense of regression-based stopping. Section 4.5 measures what this costs: it varies the demand correlation from 0 to 0.9, adds a multiplicative day factor, and tests the monotonicity of the estimated continuation value directly. Assumption 2 holds by construction of ~~(4)~~–(5)(5), since both the remaining distance and the number of downstream customers fall along the route. By finite-horizon dynamic programming, define value functions backwards from $V_m(w) \equiv 0$ on $\{w \leq B\}$ by

$$C_k(w) = \mathbb{E}[E_{k+1} \mathbf{1}\{W_{k+1} > B\} + V_{k+1}(W_{k+1}) \mathbf{1}\{W_{k+1} \leq B\} | W_k = w], \quad (9)$$

$$V_k(w) = \min\{H_k, C_k(w)\}, \quad k = m-1, \dots, 1, \quad (10)$$

where $C_k(w)$ is the expected cost of *continuing past stop k and playing optimally thereafter*. The optimal policy is the first-passage rule

$$\sigma^* = \min\{k < m : C_k(W_k) > H_k\} \quad (11)$$

by the standard verification argument for finite-horizon stopping problems (Chow et al., 1971; Peskir and Shiryaev, 2006). Everything difficult about the problem is therefore statistical: the demand law is unknown, so C_k must be estimated from historical load paths, and the quality of the policy is the quality of that estimate in the region where the comparison (11) is close.

3.3. Two structural defects of the incumbent approach

The system this work replaces—like much of the rule-based literature—estimated a per-stop breach *probability* $p_k(w)$ by isotonic regression of a binary label on W_k across historical routes and triggered a handoff when

$\hat{p}_k(W_k) > \tau$ for a tuned constant τ . Both design choices fail structurally, not statistically, and both failures were observed in the field before they were understood formally.

The first defect concerns the label. The natural per-route summary in logged data is the endpoint total W_m , and the incumbent labelled a route as overflowing by $Y^{\text{end}} = \mathbf{1}\{W_m > B\}$ rather than by the physical event $Y^{\text{peak}} = \mathbf{1}\{\max_{k \leq m} W_k > B\}$. We state the consequence formally; despite its simplicity, it explains a genuine field failure in which a deployed policy saved exactly nothing, and the same endpoint-for-peak substitution recurs whenever monitoring data are aggregated per route.

Proposition 1 (Endpoint-label bias). *For every path law and every B , $\{W_m > B\} \subseteq \{\max_{k \leq m} W_k > B\}$, hence $\mathbb{P}(Y^{\text{end}}=1) \leq \mathbb{P}(Y^{\text{peak}}=1)$, and the endpoint-trained regression estimates a conditional probability of the wrong event. The gap is extremal on collect-then-deliver routes: if the route structure forces ~~$W_m = 0$~~ $W_m \leq 0$ almost surely and $B > 0$, then $Y^{\text{end}} = 0$ a.s. while $\mathbb{P}(T \leq m)$ may be arbitrary in $[0, 1)$. In that regime ~~, under flat prices $(\omega_F, C_{\text{fail}})$, any every~~ threshold policy trained on endpoint labels ~~converges to the never-handoff policy with expected cost $C_{\text{fail}} \mathbb{P}(T \leq m)$, whereas the stopping rule (11) attains at most $\omega_F \mathbb{P}(2 \leq T \leq m) + C_{\text{fail}} \mathbb{P}(T = 1)$. Whenever breaches occur but rarely at the first stop, the endpoint-trained policy therefore inflates execution cost by approximately the factor C_{fail}/ω_F~~ coincides with the reactive policy and forgoes the entire value of preventive recourse, $\Delta := V^{\text{react}} - V^*$, where $V^{\text{react}} = \mathbb{E}[E_T \mathbf{1}\{T \leq m\}]$ and $V^* = \inf_{\sigma} \mathbb{E}J(\sigma)$. Under flat prices $(\omega_F, C_{\text{fail}})$ with $0 \leq \omega_F \leq C_{\text{fail}}$ and routes with $m \geq 2$ stops (so that at least one decision epoch precedes the last stop),*

$$\begin{aligned} & \max\{0, (C_{\text{fail}} - \omega_F) \mathbb{P}(2 \leq T \leq m) - \omega_F \mathbb{P}(T = \infty)\} \\ & \leq \Delta \leq (C_{\text{fail}} - \omega_F) \mathbb{P}(2 \leq T \leq m), \quad (12) \end{aligned}$$

and the upper bound is attained when the breach is announced one stop ahead, that is, when $T - 1$ is a stopping time on $\{2 \leq T \leq m\}$.

Proof. The inclusion is immediate since $W_m \leq \max_{k \leq m} W_k$. If ~~$W_m = 0$~~ $W_m \leq 0$ a.s. and $B > 0$ then $Y^{\text{end}} = 0$ a.s., so any regression of Y^{end} on any covariate is the zero function on the support of the data; a trigger $\hat{p}_k(W_k) > \tau$ with $\tau > 0$ never fires, and the policy ~~pays $E_T = C_{\text{fail}}$ exactly on $\{T \leq m\}$~~ is the reactive one. The reactive policy is feasible, so $\Delta \geq 0$. For the upper bound ~~on the optimum, rule (11) is optimal and therefore no worse than the clairvoyant policy of §3.6 restricted to information $\mathcal{F}_k$: on $\{2 \leq T \leq m\}$ a handoff at $T - 1$ costs ω_F , every policy adapted to $(\mathcal{F}_k)$ pays C_{fail} on $\{T = 1\}$, where~~ no decision epoch precedes the breach, ~~and at least $\min\{\omega_F, C_{\text{fail}}\} = \omega_F$ on $\{T = \infty\}$ continuing forever costs $\{2 \leq T \leq m\}$, and at least 0.~~ Taking expectations gives the bound, ~~and the ratio of the two costs tends to C_{fail}/ω_F as $\mathbb{P}(T=1) \rightarrow 0$ on $\{T = \infty\}$; hence $V^* \geq C_{\text{fail}}\mathbb{P}(T=1) + \omega_F \mathbb{P}(2 \leq T \leq m)$,~~ and subtracting from $V^{\text{react}} = C_{\text{fail}}\mathbb{P}(T \leq m)$ gives the right-hand inequality. It holds with equality when handing off at $T - 1$ is an admissible policy, which is the case exactly when $T - 1$ is a stopping time. For the lower bound, the policy “hand off after stop 1 unless the route has already breached” is admissible and costs $C_{\text{fail}}\mathbb{P}(T=1) + \omega_F \mathbb{P}(T \geq 2)$, the event $\{T \geq 2\}$ including $\{T = \infty\}$; subtracting its cost from V^{react} bounds Δ from below. $\square$

The bounds (12) say that when breaches are frequent relative to clean days, the endpoint label costs the operator almost the whole value of the handoff lever, $(C_{\text{fail}} - \omega_F)$ per breach that could have been anticipated. The synthetic collect-then-deliver scenario of §4.2 realises this regime: the endpoint-trained policy saves exactly nothing there. BATON therefore labels and calibrates on the running peak everywhere; in particular, when the slack must be calibrated empirically, it uses the $(1-\alpha)$ -quantile of historical *peaks*, never of endpoint totals.

The second defect concerns the trigger. Even with the label corrected, a ~~fixed probability threshold~~ myopic comparison is the wrong functional form,

and its error has a sign:-

~~Take flat prices $H_k \equiv \omega_F$, $E_k \equiv C_{\text{fail}}$, and let. Write $C_k^0(w) = \mathbb{E}[E_T \mathbf{1}\{T \leq m\} \mid W_k = w, T > k]$ for the cost of continuing from (k, w) and never acting again. The myopic rule hands off after stop k when $C_k^0(W_k) > H_k$. Under flat prices $C_k^0(w) = C_{\text{fail}} p_k(w)$ with $p_k(w) = \mathbb{P}(T \leq m \mid W_k = w, T > k)$ be the true conditional the peak-overflow probability. The myopic rule “hand off iff $C_{\text{fail}} p_k(w) > \omega_F$ ”, i.e., and the myopic rule is exactly the probability threshold $p_k(w) > \tau$ with $\tau = \omega_F / C_{\text{fail}}$, has stopping region containing that.~~

Proposition 2 (Myopic over-triggering). *Under Assumption 1, at every stop k the stopping region of the optimal rule (11) at every stop:- (11) is contained in that of the myopic rule:*

$$\{w : C_k(w) > \underline{\omega_F} H_k\} \subseteq \{w : C_{\text{fail}} p_k^0(w) > \underline{\omega_F} H_k\}.$$

~~The Moreover $C_k(w) < C_k^0(w)$ whenever, with positive probability given $W_k = w$ and $T > k$, the path survives to some stop $j \in \{k+1, \dots, m-1\}$ at which the myopic rule itself fires. If C_k and C_k^0 are continuous and this condition holds at the myopic boundary $\bar{w}_k = \inf\{w : C_k^0(w) > H_k\} < B$, the inclusion is strict at stop k whenever $\mathbb{P}(T > k+1 \mid W_k = w, T > k) > 0$ on a set of w of positive measure—that is, whenever the breach, if it comes, may still be more than one stop away: a right-neighbourhood of $\bar{w}_k$ is stopped by the myopic rule but continued by the optimal one.~~

Proof. “Continue and never act again” is ~~a feasible continuation policy an~~ admissible continuation from (k, w) with expected cost ~~from state (k, w)~~ exactly $C_{\text{fail}} p_k(w) C_k^0(w)$; since $C_k(w)$ is the infimum over ~~all continuation policies~~, $C_k(w) \leq C_{\text{fail}} p_k(w)$, giving admissible continuations, $C_k \leq C_k^0$, which gives the inclusion. For strictness, split $p_k(w) = q_1(w) + q_2(w)$ with $q_1(w) = \mathbb{P}(T = k+1 \mid \cdot)$ and $q_2(w) = \mathbb{P}(k+1 < T \leq m \mid \cdot)$ the strict inequality, let ρ be the first stop $j > k$ with $T > j$ and $C_j^0(W_j) > H_j$, and consider the ~~feasible policy “continue one stop; if no breach, hand off at $k+1$ ” restricted to the event driving q_2 .~~

~~Its cost~~ continuation that hands off at ρ and never acts otherwise. It differs from “never act” only on $A = \{\rho < m\}$, where it pays H_ρ while never acting costs, conditionally on $\mathcal{F}_\rho$ and by the strong Markov property, $C_\rho^0(W_\rho) > H_\rho$. If $\mathbb{P}(A \mid W_k = w, T > k) > 0$, its expected cost is strictly below $C_k^0(w)$, and so is $C_k(w)$. Finally, C_k^0 is non-decreasing (it is the continuation value of the problem with $H \equiv \infty$, to which Proposition 4 applies), so $C_k^0 > H_k$ on $(\bar{w}_k, B]$ and, by continuity, $C_k^0(\bar{w}_k) = H_k$. Then $C_k(\bar{w}_k) < C_k^0(\bar{w}_k) = H_k$, and continuity of C_k gives $C_k < H_k < C_k^0$ on a right-neighbourhood of $\bar{w}_k$. $\square$

Proposition 2 is qualitative. The next result prices the over-triggering. Let σ^0 and σ^* be the stopping times of the myopic and the optimal rule, and write $L_k := 0$ on $\{T = \infty\}$ and $L_k := \min\{\min_{k < j < T} H_j, E_T\}$ on $\{k < T < m\}$ for the clairvoyant cost of the remainder of the day after stop k : the cheapest later handoff before the breach, or the breach itself.

Proposition 3 (The price of over-triggering). *Under Assumption 1, $\sigma^0 \leq \sigma^*$ almost surely, and*

$$\begin{aligned} 0 \leq \mathbb{E} J(\sigma^0) - \mathbb{E} J(\sigma^*) &= \mathbb{E}[(H_{\sigma^0} - C_{\sigma^0}(W_{\sigma^0}))\mathbf{1}\{\sigma^0 < \sigma^*\}] \\ &\leq \mathbb{E}[(H_{\sigma^0} - L_{\sigma^0})\mathbf{1}\{\sigma^0 < \sigma^*\}]. \end{aligned} \quad (13)$$

Under flat prices the right-hand side is at most $C_{\text{fail}}q_1(w) + \omega_F q_2(w)$, strictly below $C_{\text{fail}}(q_1(w) + q_2(w)) = C_{\text{fail}}p_k(w)$ whenever $q_2(w) > 0$ $\omega_F \mathbb{P}(\sigma^0 < \sigma^, T = \infty) \leq \omega_F \mathbb{P}(\sigma^0 < \infty)$*

Proof. If the optimal rule stops at k then W_k lies in its stopping region, hence in the myopic one by Proposition 2; the myopic rule has therefore stopped at k or earlier, so $\sigma^0 \leq \sigma^*$. On $\{\sigma^0 = \sigma^*\}$ the two policies take identical actions all day and pay the same. On $\{\sigma^0 < \sigma^*\}$, an event in $\mathcal{F}_{\sigma^0}$, the myopic rule pays H_{σ^0} while the optimal rule continues from (σ^0, W_{σ^0}) , whose conditional expected cost is $C_{\sigma^0}(W_{\sigma^0})$ by the strong Markov property. This gives the identity, and the difference is non-negative because $\omega_F < C_{\text{fail}}$.

Hence $C_k(w) < C_{\text{fail}} p_k(w)$ there, and every state with σ^* is optimal. For the inequality, every admissible continuation from stop k pays, path by path, either some H_j with $k < j < T$, or E_T , or nothing on a clean day, hence at least L_k ; taking conditional expectations, $C_k(W_k) \geq \mathbb{E}[L_k | \mathcal{F}_k]$ on $\{T > k\}$, and the tower property gives the bound. Under flat prices $H_{\sigma^0} - L_{\sigma^0}$ equals ω_F between the two values is myopically stopped but optimally continued, on $\{T = \infty\}$, 0 on $\{T \geq \sigma^0 + 2\}$ and $\omega_F - C_{\text{fail}} < 0$ on $\{T = \sigma^0 + 1\}$. $\square$

Proposition 3 turns the critique into an accounting statement. The premium a threshold rule pays for its simplicity is at most the handoff price times the probability that it intervenes on a day that would have ended cleanly, plus, under declining prices, the saving forgone by handing off before a cheaper later opportunity. A threshold is therefore “good enough” exactly when both terms are small: when the states in which it fires lead to a breach almost surely, as on tightly packed routes whose risk sits in the next stop or two, and when prices decline slowly along the route. It is poorer on plans that carry slack, where the rule fires on more days that would have recovered by themselves. Section 4.4 evaluates both sides of (13) on our benchmarks.

Proposition 2 explains a second also explains a field observation: tuned thresholds drifted upward—toward less triggering—precisely on long routes, chasing option value the functional form cannot express. Under state-dependent prices the failure becomes qualitative rather than quantitative is also one of representation: the comparison $p_k(w) E > H_k - C_k^0(w) > H_k$ with per-stop H_k is not even representable as one a single threshold on p_k , whereas the cost comparison of (11) is unchanged. A threshold that varies by stop can represent the optimal boundary—since C_k is monotone (Proposition 4), the optimal stopping region at each stop is an upper set $\{w > w_k^*\}$ —but its $m - 1$ boundaries must then be searched directly against realised cost; §4.1 includes such a position-dependent threshold, and a cost-scaled myopic rule, among the competitors, to isolate the value of *estimating* the boundary by backward induction rather than searching for it.

3.4. Estimating continuation values from historical paths

Estimating (9) requires a conditional expectation of future costs *under optimal future play*—precisely the structure of American-option pricing by simulation, and we adopt the regression-based backward induction of Longstaff and Schwartz (2001). Sweep the N historical paths backwards, maintaining for each path s the realised future cost ϕ_s under the policy already constructed for later stops, initialised by $\phi_s = E_{T_s}$ if path s breaches and 0 otherwise; at stop k , regress ϕ on W_k among the paths still alive (those with $T_s > k$) to obtain $\hat{C}_k$; then apply the implied action, replacing ϕ_s by H_k on paths where $\hat{C}_k(W_{k,s}) > H_k$, so that stop $k - 1$ sees costs under the improved later policy. The regression target is a cost supported on $\{0\} \cup \{H_j\}_{j>k} \cup \{E_j\}_{j>k}$, not a probability, and the estimator we use for every regression is *isotonic* least squares (Barlow et al., 1972). The shape constraint is not a convenience; it is a theorem about the estimand:

Proposition 4 (Monotone continuation value). *Under Assumptions 1 and 2, for every $k \in \{1, \dots, m - 1\}$ the continuation value $C_k(\cdot)$ of (9) is non-decreasing on $(-\infty, B]$. The same holds for any recourse menu whose actions either end the route at a price known at stop k or reset the load to a fixed level at a known price—in particular for the three-action recursion (14) below, and for the menu without the handoff lever, whose continuation value is C_k^0 .*

Proof. Backward induction on k . Write the integrand of (9) as $\varphi_{k+1}(x) = E_{k+1} \mathbf{1}\{x > B\} + V_{k+1}(x) \mathbf{1}\{x \leq B\}$, so that ~~by $C_k(w) = \mathbb{E}[\varphi_{k+1}(W_{k+1}) | W_k = w]$.~~ By Assumption 1 $C_k(w) = \mathbb{E}[\varphi_{k+1}(w + g_{k+1})$ with g_{k+1} independent of the conditional law of W_{k+1} is stochastically increasing in w ; it suffices that, ~~so C_k is non-decreasing as soon as φ_{k+1} be non-decreasing, monotonicity being preserved by the shift $w \mapsto w + g_{k+1}$ pathwise and by expectation.~~ is. At $k + 1 = m$ we have $V_m \equiv 0 \leq E_m$, so φ_m is a non-decreasing step function. For $k + 1 < m$, the inductive hypothesis makes C_{k+1} non-decreasing, ~~hence $V_{k+1} = \min\{H_{k+1}, C_{k+1}\}$ is and V_{k+1} is the minimum of C_{k+1} and of constants~~

(the handoff price, and the return price plus the fresh-start value), hence non-decreasing on $(-\infty, B]$ with $V_{k+1} \leq H_{k+1} \leq E_{k+1}$. Moreover $V_{k+1}(x) \leq C_{k+1}(x) \leq E_{k+1}$: the cost of continuing and never acting again is at most the largest emergency price still reachable, $\max_{j \geq k+2} E_j$, which is at most E_{k+1} by Assumption 2. For $x \leq x'$: if both exceed B , $\varphi_{k+1}(x) = \varphi_{k+1}(x') = E_{k+1}$; if both are at most B , $\varphi_{k+1}(x) = V_{k+1}(x) \leq V_{k+1}(x') = \varphi_{k+1}(x')$; if $x \leq B < x'$, then $\varphi_{k+1}(x) = V_{k+1}(x) \leq E_{k+1} = \varphi_{k+1}(x')$. Thus φ_{k+1} is non-decreasing in every case, completing the induction. No ordering between H_k (or R_k) and E_k is used. $\square$

Isotonic regression is therefore the least-squares projection onto exactly the function class known to contain the truth; it requires no basis choice, bandwidth, or regularisation parameter, and it inherits classical consistency guarantees (Ayer et al., 1955; Barlow et al., 1972). Convergence of the full backward-induction scheme to the optimal stopping value as $N \rightarrow \infty$ follows the analysis of Clément et al. (2002), whose argument requires only that each regression step be an L^2 -consistent estimator within the approximating class; ~~monotone functions on a compact range~~ bounded monotone functions form a Glivenko–Cantelli class, so the isotonic step qualifies whenever the truth is monotone, which Proposition 4 guarantees under Assumption 1. When the assumption holds only approximately, the isotonic fit is the monotone projection of the conditional expectation given W_k ; §4.5 checks directly, on fifty thousand paths per route, whether that conditional expectation is monotone under strongly dependent demand.

Remark 1. ~~For very small histories ($N \lesssim 500$ paths) backward induction amplifies estimation noise, because errors at late stops propagate. Backward induction propagates estimation error from late stops into the regression targets of early stops. BATON then falls,~~ so its advantage over a directly tuned threshold should shrink as the history shortens. The submitted version of this paper therefore recommended falling back to the label-corrected ~~threshold variant—~~isotonic estimation of the peak-overflow probability with one threshold

~~tuned on realised training cost—which inherits the label fix but not the option-value correction. Its empirical price is quantified in §4~~tuned threshold below about 500 training paths. The data-budget experiment of §4.7 does not support that recommendation: BATON beats the tuned threshold at every history length we tested, down to 100 training days per route. The isotonic projection pools information along the load axis, which keeps the propagated error small. The regime in which backward induction does over-fit is a different one—breaches so rare that a history contains only a few dozen of them (§4.5). There the full policy learns to decline acting, while its handoff-only restriction still over-fits and loses a little money.

3.5. Pricing the full action set

~~The recursion (9)–(10) extends to the depot-return lever with one additional term~~

The depot-return transition. After stop k the vehicle carries $L_0 + W_k$, where L_0 is the departure load. A return ~~after stop k resets the running sum to zero—collected pickups leave the vehicle, and drives to the remaining deliveries are reloaded to plan—at depot,~~ unloads every pickup collected so far, and sends the vehicle back to stop $k + 1$ to continue its route in the planned order, at the price R_k of (6). The policy records the post-return state as $W = 0$ against the unchanged slack B , that is, as if the vehicle left the depot again with its departure load L_0 . This is a conservative convention. Unloading the pickups leaves on board exactly the undelivered cargo $L_0 - D_{\leq k}$, where $D_{\leq k} = \sum_{j \leq k} d_j$ is the volume already delivered, so the exact post-return state is $x_k = -D_{\leq k} \leq 0$; the convention understates the residual capacity after a return by $D_{\leq k}$ and, if anything, undervalues the lever. We adopt it because it fixes the post-return state, which keeps the fresh-start value below a single number per stop; with the exact reset it becomes a function of the observed delivered volume. Section 4.5 prices the exact reset and shows that it raises the saving further, so the results

we report for the depot return are conservative. Returns may be repeated: after a return the vehicle re-enters the same decision process in the state $(k, 0)$. ~~The three-action Bellman recursion is therefore,~~ so further returns and a later handoff remain available. (The published restocking competitor is restricted to a single return, as is usual in that literature.)

The recursion (9)–(10) therefore extends to the depot-return lever with one additional term:

$$V_k(w) = \min\{C_k(w), H_k, R_k + C_k(0)\}, \quad k = m - 1, \dots, 1, \quad (14)$$

with C_k still defined by (9); the quantity $F_k := C_k(0)$ is the *fresh-start value*, the expected cost of the remainder of the day from the reset state under optimal play with the full action set, including any further returns or a later handoff. The optimal policy exercises, at each stop, the cheapest action whose price is below the estimated cost of continuing.

~~Estimating (14) requires care on one point.~~

Estimating the fresh-start value. The isotonic estimate $\hat{C}_k$ is fitted on the *alive* paths at stop k , whose load states concentrate away from zero on routes that depart nearly full; ~~naively~~ reading off $\hat{C}_k(0)$ ~~then extrapolates~~ would extrapolate the monotone fit to the edge of its support, where it is biased downward, and the fitted policy ~~over-returns—a failure mode we observed and document in §4~~ would over-return. BATON instead evaluates F_k ~~exactly with respect to its own fitted policy~~ with respect to its own fitted policy: during the backward sweep, when stop k is reached, the models and fresh-start values of all later stops are already available, so F_k is computed by simulating the reset state $(k, 0)$ forward through the training paths under the already-fitted downstream policy, nested returns valued by the already-computed F_j , $j > k$. This adds one vectorised forward pass per stop— $O(Nm)$ work—and ~~removes the extrapolation bias entirely~~ avoids the extrapolation.

~~One further safeguard completes the policy~~The resulting estimate $\hat{F}_k$ is subject to three biases of different signs. (i) The downstream policy is the fitted, not the optimal, one, which pushes $\hat{F}_k$ above the optimal F_k in population; this is also the value the deployed policy will actually realise after a return, so it is the consistent price for the policy that is deployed. (ii) The forward pass runs on the paths on which the downstream models were fitted, which pushes $\hat{F}_k$ down by the usual in-sample optimism. (iii) When demand is positively dependent, the suffix after a return is correlated with the state that triggered it: returns are chosen on heavy days, whose remainder tends to be heavy too, so an unconditional F_k under-prices the return on exactly the days on which it is used. Section 4.5 measures all three on fifty thousand independent paths per route: (i) and (ii) nearly cancel, while (iii) is material on delivery-dominated urban routes. The third bias has a direct remedy that stays within the method. Regressing each training path’s realised post-return cost, computed from that path’s own future increments, on its pre-return state W_k gives a *state-conditional* fresh-start value $\hat{F}_k(W_k)$ at the cost of one more isotonic fit per stop. It is monotone for the same reason as C_k , and it is flat, so that the policy reduces to BATON, when increments are independent. We report this variant as BATON-CF.

Deployment selection. On routes where the return lever cannot pay—remote depots, hostile geometry—including it in the menu adds estimation noise without adding value, and ~~a noisy extra action can degrade a the~~ finite-sample ~~policy~~biases above can then make the richer menu slightly worse than ~~the handoff-only one~~. BATON therefore ~~performs deployment selection: it~~ fits both the two-action and the three-action policy on the training paths, evaluates both on those same ~~training~~ paths, and deploys whichever is cheaper. The selection uses no test data ~~, and~~ costs one simulation pass ~~, and guarantees that the richer menu is used exactly where the data say it helps~~. Because it compares in-sample costs it inherits bias (ii), which is why it is a safeguard rather than a guarantee; BATON-CF, which removes bias (iii) at its source,

Algorithm 1 BATON offline phase: peak-aware backward induction over the full action set

Require: historical increments $g_k^{(s)}$, $s = 1..N$, $k = 1..m$; slack B ; price schedules (H_k) , (E_k) , (R_k)

- 1: $W_k^{(s)} \leftarrow \sum_{j \leq k} g_j^{(s)}$; $T_s \leftarrow \min\{k : W_k^{(s)} > B\}$ (∞ if none) $\triangleright$ peak, not endpoint
- 2: $\phi_s \leftarrow E_{T_s} \mathbf{1}\{T_s \leq m\}$ $\triangleright$ cost under “never act”
- 3: **for** $k = m - 1$ **down to** 1 **do**
- 4: $A \leftarrow \{s : T_s > k\}$ $\triangleright$ paths alive at stop k
- 5: $\hat{C}_k \leftarrow$ isotonic fit of $\{\phi_s\}_{s \in A}$ on $\{W_k^{(s)}\}_{s \in A}$
- 6: $F_k \leftarrow$ mean cost of simulating state $(k, 0)$ forward through $\{\hat{C}_j, F_j\}_{j > k}$
 $\triangleright$ exact fresh-start value
- 7: **for** $s \in A$: $\phi_s \leftarrow \min\{\hat{C}_k(W_k^{(s)}), H_k, R_k + F_k\}$ if an action is cheaper
 $\triangleright$ policy improvement
- 8: **end for**
- 9: **return** $\{\hat{C}_k, F_k\}_{k=1}^{m-1}$

performs as well without it (§4.5).

3.6. Algorithms, complexity, and reference bounds

Algorithm 1 states the offline phase for the full action set; Algorithm 2 the online loop, whose logic is also drawn in Figure 1. The offline phase performs one isotonic fit and one fresh-start evaluation per stop: $O(N \log N + Nm)$ per stop, hence $O(mN(\log N + m))$ per route in ~~total—single-digit milliseconds at total.~~ Measured on one core (§4.7), fitting the full policy with deployment selection takes 27 ms per route at $N = 10^3$ and route lengths up to $m \approx 25$ on one-commodity-core (median over our routes), which is what makes nightly re-fitting across a metropolitan fleet unremarkable. The online decision is one ~~monotone-function-evaluation-and-evaluation of a monotone step function—a binary search over its breakpoints—and~~ two scalar comparisons per stop, about 1 μ s per decision.

~~Two-Three~~ reference points calibrate the empirical results. The *clairvoyant bound* pays 0 on $\{T = \infty\}$ and $\min\{\min_{k < j} H_k, E_j\}$ on $\{T = j\}$ —the

Algorithm 2 BATON online phase: priced-action trigger

Require: $m, B, (H_k), (E_k), (R_k)$, fitted $\{\hat{C}_k, F_k\}$

```
1:  $W \leftarrow 0$ 
2: for  $k = 1$  to  $m$  do
3:   serve stop  $k$ ; observe  $d_k, p_k$ ;  $W \leftarrow W + p_k - d_k$ 
4:   if  $W > B$  then return (EMERGENCY,  $E_k$ )
5:   end if ▷ physical guard: running peak
6:   if  $k = m$  then return (COMPLETE, 0)
7:   end if
8:    $c \leftarrow \hat{C}_k(W)$ 
9:   if  $H_k < c$  and  $H_k \leq R_k + F_k$  then return (HANDOFF,  $H_k$ )
10:  end if
11:  if  $R_k + F_k < c$  then
12:    pay  $R_k$ ;  $W \leftarrow 0$  ▷ depot return: pickups emptied, deliveries
13:    reloaded
14:  end if
15: end for
```

cheapest of any pre-breach handoff or the breach itself, with a breach at the first stop leaving no decision epoch; no policy adapted to $(\mathcal{F}_k)$ that is restricted to the handoff lever pays less on any path. Because it is defined for the handoff lever, a policy with the richer menu may legitimately beat it, ~~and in our experiments~~: a depot return costs less than a handoff and does not end the route, so it can prevent a breach at a price no handoff-only policy, clairvoyant or not, can match. When BATON ~~does—the cleanest evidence that the action generalisation has real value.~~ The exceeds this bound, the excess measures the value of the second lever, not a statistical artefact. The plug-in dynamic ~~program~~programs replaces—replace the isotonic step by quantile-binned conditional means; ~~run.~~ Run with fifty times BATON’s training data it approaches—they approach the exact solution of the stopping problem ~~and serves as a~~ in the (k, W_k) state and serve as near-exact ~~yardstick,~~ while—yardsticks, one for the handoff-only problem (DP_{50k}) and one for the three-action problem (DP_{50k}^3 , whose fresh-start values are computed on an

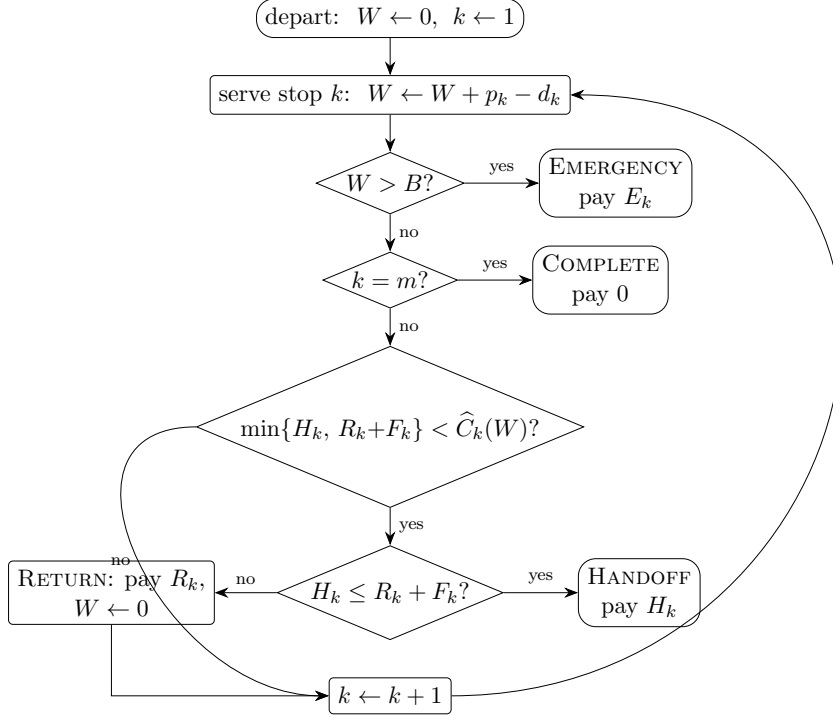


Figure 1: Online decision loop of BATON. After every stop the policy compares the estimated cost of continuing against the current prices of the two recourse levers and exercises the cheaper lever only when continuing is dearer; a depot return resets the load state and the loop proceeds on the same route.

independent half of the sample so that they carry no in-sample optimism); at equal data ~~it~~ the handoff-only program is a fair equal-information competitor (DP_N). We emphasise a point of method: mixed-integer programming, the exact technology of the *planning* stage, cannot compactly encode a non-anticipative multistage stopping rule, so backward dynamic programming is the correct exact anchor for the *execution* stage; we use MIP where it belongs, certifying the routing plans beneath all our experiments (§4.3).

~~Figure 2 makes~~ Figures 2 and 3 make the machinery concrete on one route of the Hanoi instance introduced below: the fan of simulated load paths with a demand-spike day highlighted, the fitted continuation cost crossing the handoff price at stop 2—ten stops before the breach the reactive policy

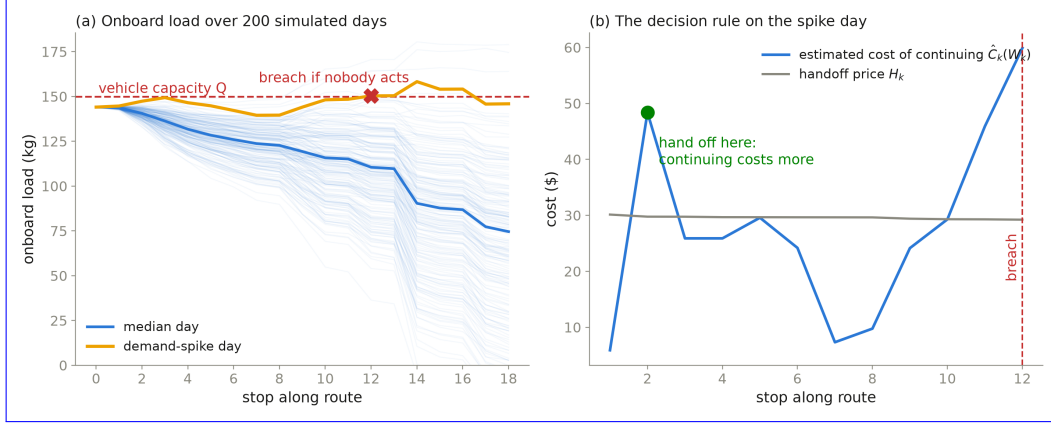


Figure 2: BATON on one 18-stop route in Hanoi. (a) Onboard load across 200 simulated days; the highlighted day breaches capacity at stop 12 if nobody acts. (b) The decision rule on that day: the estimated cost of continuing $\hat{C}_k(W_k)$ crosses the per-stop handoff price at stop 2, where the policy hands off.

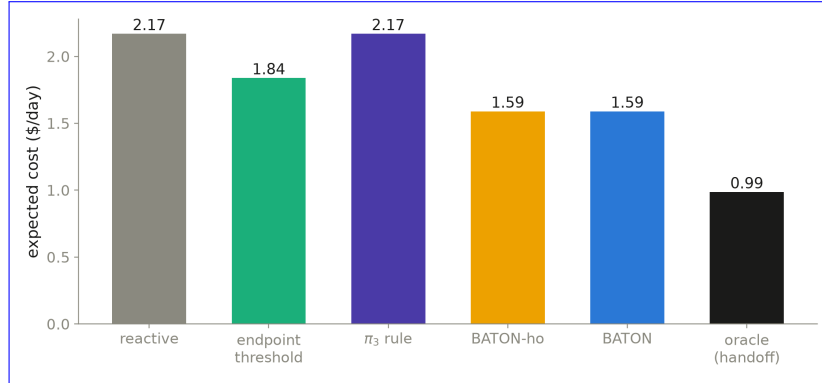


Figure 3: Expected execution cost per policy on the route of Figure 2, over 2,000 held-out days. On this route deployment selection keeps the depot return switched off, so BATON coincides with BATON-HO; the oracle is the handoff-only clairvoyant bound.

suffers—and the resulting ~~cost distribution~~ expected cost over two thousand held-out days.

4. Computational study

Our computational study is designed around a simple discipline: every claim about the policy must survive a change of environment, a change of

competitor, and a change of yardstick. This section first describes the environments, competitors, and protocol (§4.1), then the headline comparison across planning regimes [with its statistical analysis and tail risk](#) (§4.2), the large-scale and real-map benchmarks with the planning-stage certification (§4.3), the comparison against learning methods ~~together with two deliberate negative results~~ [and the empirical price of over-triggering](#) (§4.4), [the robustness of the conclusions to the modelling assumptions](#) (§4.5), [shared standby capacity](#) (§4.6), [data and computing budgets](#) (§4.7), and finally the economic and spatial sensitivity analyses (§4.8). All experiments, result files, and figures are reproducible from a single repository, and every table in this paper is generated mechanically from the result files.

4.1. Instances, planners, competitors, and protocol

Instances. Three families are used. The 40 Dethloff instances ([Dethloff, 2001](#)), 50 customers each, are the classical simultaneous pickup-and-delivery benchmark. The Salhi–Nagy family ([Salhi and Nagy, 1999](#)), 14 instances of 50–199 customers, is derived from the CMT instances ([Christofides et al., 1979](#)) by the standard coordinate-ratio demand split and extends the size range. Finally, we introduce a set of instances built on the real drive networks of Ho Chi Minh City, Hanoi, New York, Paris, and Shanghai (6 km radius around the centre, 5,800–15,800 road nodes per city): customers are placed at *real shop locations* drawn from OpenStreetMap’s retail layer, snapped to the nearest network node, so the spatial demand distribution is the city’s actual one—market streets, malls, commercial clusters—and distances are true road distances ([Boeing, 2017](#)). Instance sizes reach 400 customers; Figure 4 shows four of the instances with their planned routes. Demands follow the benchmark protocol (gamma marginals, coefficient of variation 0.3, equicorrelated Gaussian copula $\rho = 0.6$ [among the deliveries and, independently, among the pickups](#)); the city instances use parcel-van magnitudes, with mean deliveries of 8 kg, pickup fractions uniform on $[0.2, 0.8]$, and 150 kg vehicles, which yields the long 12–25-stop routes where execution policy matters most.



Figure 4: City instances on real street networks, with customers at real shop locations (Hanoi, New York, Paris, Shanghai; 100 customers each; depot marked ★). Note the true retail clustering—Hanoi’s Old Quarter, Manhattan’s commercial spine—that a uniform spatial convention misses; §4.8 quantifies its effect.

Because deliver-only customers are common in practice, every city instance also has two deliver-only twins in which the pickup of a random 25% or 50% of the customers is set to zero; the twins are re-planned.

Planning regimes. Routes are constructed by an adaptive large neighbourhood search planner running under six capacity-feasibility gates of increasing conservatism: a deterministic gate on the nominal load peak (DET); a sample-average CVaR gate (SAA); a Wasserstein distributionally robust gate (WDRO) (Mohajerin Esfahani and Kuhn, 2018); the quadrant-budget robust gate of Gounaris et al. (2013) (ROB-G); a budget-uncertainty gate in the sense of Bertsimas and Sim (2004) (ROB-BS); and a moment-ambiguity gate in the closed form implied by Ghosal et al. (2024) (M-DRO). With a type-1 Wasserstein ball over unbounded support and a peak-load loss that is 1-Lipschitz, the worst-case CVaR equals the empirical CVaR plus the constant $\varepsilon/(1 - \alpha)$ (Mohajerin Esfahani and Kuhn, 2018), so the WDRO gate is the SAA gate at a tighter threshold; we retain it as a distinct, more conservative planning regime rather than as an independent modelling paradigm. Every execution policy is evaluated on the identical plans of every gate, so planning quality never contaminates the policy comparison; the plans themselves are certified in §4.3.

Competitors. ~~Eleven~~ Fourteen execution policies are compared, all trained (where applicable) on scenarios strictly disjoint from the test set: the *reactive* policy that never intervenes; the incumbent endpoint-labelled threshold and its myopic variant, retained as ablations of Propositions 1 and 2; the *label-corrected tuned threshold* (peak labels, one threshold tuned on realised training ~~cost—the strongest member of the threshold family); cost~~); a position-dependent threshold with one load threshold per stop, tuned by coordinate descent on realised training cost, which can represent any monotone stopping boundary and therefore isolates the value of estimating that boundary by backward induction rather than searching for it; the three published rule-based restocking triggers π_1, π_2, π_3 of Salavati-Khoshghalb et al. (2019), adapted to the handoff lever and grid-tuned on realised cost; a *rollout* policy in the sense of Secomandi (2001), one step of policy improvement over the reactive ~~base~~base—which, under position-dependent prices, is exactly

the myopic rule $C_k^0(W_k) > H_k$ —and a *cost-scaled rollout* that hands off when $C_k^0(W_k) > \theta H_k$ with θ tuned on realised cost; a *depot-return restocking* policy in the tradition of Yang et al. (2000) and Florio et al. (2020), with real detour prices and a tuned trigger; BATON-HO, the handoff-only restriction of our policy; BATON itself; and the equal-data plug-in dynamic program DP_N ; and the near-exact DP_{50k} . BATON-CF, the variant with a state-conditional fresh-start value, is reported in §4.5. A reinforcement-learning policy re-implemented from Iklassov et al. (2024) is compared separately in §4.4. The two near-exact programs DP_{50k} and DP_{50k}^3 and the clairvoyant bound closes every table close every table as reference points. Table 1 collects the full-roster with sources; in every results table the reported quantity is the expected-recourse saving relative to the reactive policy, in percent, so *higher is always better*.

Protocol. Per route, policies are fitted on For each instance, one set of 10^3 training scenarios and evaluated on days, one set of 2×10^3 independent test days; results aggregate to planlevel and then across instances. All significance statements are paired test days and one set of 5×10^4 reference days are drawn from independent random streams whose seeds are fixed functions of the instance name, so every run of the pipeline draws identical scenarios. All routes of a plan share the same days, as they would in operation. Every data-driven policy—BATON, the thresholds, the rule-based triggers, the rollouts, the restocking trigger, DP_N and the RL policy—sees exactly the same 10^3 training days; hyper-parameters of the tuned competitors are selected on realised cost over those same training days (there is no separate validation set, which, if anything, favours the tuned competitors: BATON has nothing to tune). The test days are used only for the reported evaluation. The reference programs use the 5×10^4 reference days: DP_{50k}^3 fits on one half and values its fresh-start states on the other. Offline fitting time and online decision time are reported separately in §4.7.

Table 1: Execution policies compared in this paper. Policies marked “tuned” receive their best hyper-parameter(s) by search on realised cost over the training scenarios; BATON has no tunable parameter. The last three rows are reference points, not competitors.

Label	Policy	Source
reactive	never intervenes; emergency on breach	baseline
endpoint + τ	endpoint-label threshold (tuned), and its myopic variant	ablation (Props. 1, 2)
thr.	peak-label threshold (tuned)	threshold ablation
thr.- k	one load threshold per stop (tuned by coordinate descent)	position-dependent threshold
$\pi_1\text{--}\pi_3$	rule-based restocking triggers (tuned)	Salavati-Khoshghalb et al. (2019)
roll.	one-step policy improvement (myopic cost rule)	Secomandi (2001)
roll.- θ	myopic cost rule with tuned price multiplier	cost-dependent threshold
restock	depot-return restocking, tuned trigger	Yang et al. (2000); Florio et al. (2020)
RL	policy-gradient network	Iklassov et al. (2024)
DP _N	plug-in dynamic program, equal data	Secomandi and Margot (2009)
BATON-HO	handoff-only restriction of our policy	this paper
BATON	priced-action optimal stopping	this paper
BATON-CF	BATON with state-conditional fresh-start value	this paper
DP _{50k} , DP _{50k} ³	plug-in dynamic program, 50× data, handoff-only / three actions	near-exact reference
oracle	clairvoyant per-day optimum (handoff-only)	bound

Statistical analysis. The experimental unit is the instance within a planning gate: policies are compared on the plan-level expected recourse cost, the sum over the plan’s routes of the mean over the 2×10^3 test days. Routes of a plan share their test days and are therefore not treated as independent observations, and the six gates re-use the same forty instances, so we analyse each gate separately ($n = 40$) rather than pooling them. For each comparison we report the mean paired difference in saving with a 95% bootstrap confidence interval, the number of instances won, the matched-pairs rank-biserial effect size, and a two-sided Wilcoxon signed-rank ~~tests-across-test~~ with Holm’s correction across all comparisons of a table. Route-level results for every policy, gate and instance are provided as supplementary data, and the tail of the daily plan-level ~~observations.—The fleet-economies bill~~ is analysed separately.

Fleet economics. The defaults are $F_{\text{plan}} = 35$, $F_{\text{sb}} = 20$, $F_{\text{ho}} = 5$, $c_{\text{tr}} = 3$, $F_{\text{emg}} = 40$, $s_{\text{ho}} = 1.2$, $s_{\text{emg}} = 2.5$, $p_{\text{late}} = 1.5$, $p_{\text{breach}} = 10$ (currency units per vehicle-day or per event; kilometre costs $c_{\text{km}} = 0.1$), calibrated to the order of magnitude of Southeast-Asian gig-logistics rates and swept in §4.8.

Software. All planning and execution code is Python 3.11.15 (NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.3); the isotonic regressions of Algorithm 1 use scikit-learn 1.9.0.1. The MIP certification of §4.3 solves the two-commodity flow formulation through PuLP 3.3.2 with Gurobi 13.0.2 as the primary solver and the open-source HiGHS 1.15.1 as an independent cross-check; both solvers run single-model with a 300-second budget per instance and default tolerances. Street networks and shop locations come from OpenStreetMap via OSMnx 2.1.0 (Boeing, 2017), and the reinforcement-learning baseline is trained with PyTorch ~~on an NVIDIA T4 GPU~~2.14 on the same CPU. Everything else—the ALNS planner, all six gates, and every execution policy of Table 1—is implemented from the cited papers in plain Python within our repository, so all competitors share the same simulator, cost model, and data.

Table 2: Expected-recourse saving over the reactive policy (% , mean over the 40 Dethloff instances) for each planning gate and execution policy under the three-class fleet cost model; *higher is better*. BATON (bold) has the highest saving of all implementable policies on every gate. thr. is the label-corrected tuned threshold, thr.- k the position-dependent threshold, roll.- θ the cost-scaled rollout. The last three columns are reference points, not competitors: DP_{50k} and DP_{50k}³ are near-exact dynamic programs for the handoff-only and the three-action problem with fifty times the training data, and the clairvoyant oracle is restricted to the handoff lever, so BATON may legitimately exceed it.

Gate	published / tuned competitors						this paper		reference points		
	π_3	roll.	roll.- θ	restock	thr.	thr.- k	BATON-HO	BATON	DP _{50k}	DP _{50k} ³	oracle
DET	17.0	20.8	24.9	5.8	24.7	25.8	26.1	27.9	26.4	29.5	40.7
SAA	12.6	8.9	18.4	43.7	16.3	15.4	20.1	53.7	22.1	56.4	43.5
WDRO	8.0	-4.2	8.9	34.7	-1.3	-0.2	10.0	41.2	13.2	47.8	42.0
ROB-G	15.7	10.1	19.1	43.6	16.7	16.7	21.0	53.2	22.9	56.3	43.3
ROB-BS	11.1	10.3	18.7	42.0	16.7	16.3	20.7	52.5	22.6	54.9	43.1
M-DRO	9.6	-1.4	10.3	37.4	0.5	-1.0	11.8	45.3	14.2	50.2	42.2

Table 3: Paired comparison of BATON with, in turn, the strongest competitor on each gate (the implementable policy with the highest mean saving in Table 2), its handoff-only restriction, and the near-exact three-action dynamic program. The experimental unit is the instance ($n = 40$ per gate; the routes of a plan share their test days and are aggregated within the plan). Δ : mean difference in saving (percentage points; positive favours BATON) with a 95% bootstrap confidence interval; W: instances on which BATON is better; r : matched-pairs rank-biserial effect size; p : two-sided Wilcoxon signed-rank test, Holm-adjusted over the 18 comparisons.

Gate	competitor	vs. strongest competitor				vs. BATON-HO				vs. DP _{50k} ³ (reference)			
		Δ [95% CI]	W	r	p	Δ [95% CI]	W	r	p	Δ [95% CI]	W	r	p
DET	thr.- k	+2.1 [+1.6, +2.6]	40	+1.00	$<10^{-4}$	+1.8 [+1.4, +2.2]	40	+1.00	$<10^{-4}$	-1.6 [-2.2, -1.2]	0	-1.00	$<10^{-4}$
SAA	restock	+10.0 [+8.5, +11.4]	40	+1.00	$<10^{-4}$	+33.5 [+32.0, +35.0]	40	+1.00	$<10^{-4}$	-2.7 [-3.3, -2.3]	0	-1.00	$<10^{-4}$
WDRO	restock	+6.5 [+3.5, +9.6]	31	+0.70	$<10^{-4}$	+31.2 [+28.6, +33.7]	40	+1.00	$<10^{-4}$	-6.6 [-8.9, -4.6]	3	-0.91	$<10^{-4}$
ROB-G	restock	+9.6 [+8.1, +11.1]	40	+1.00	$<10^{-4}$	+32.3 [+30.5, +34.0]	40	+1.00	$<10^{-4}$	-3.1 [-3.6, -2.6]	1	-1.00	$<10^{-4}$
ROB-BS	restock	+10.5 [+8.6, +12.5]	40	+1.00	$<10^{-4}$	+31.8 [+30.3, +33.3]	40	+1.00	$<10^{-4}$	-2.5 [-2.9, -2.0]	2	-0.99	$<10^{-4}$
M-DRO	restock	+7.9 [+5.1, +11.1]	34	+0.89	$<10^{-4}$	+33.5 [+30.9, +36.0]	40	+1.00	$<10^{-4}$	-5.0 [-7.2, -3.1]	5	-0.84	$<10^{-4}$

4.2. Headline comparison across planning regimes

Table 2 reports the central result, and Figure 5 draws it. ~~Three~~, and Table 3 gives the paired statistics. Four observations organise the discussion.

First, BATON ~~attains the lowest execution cost~~ has the highest saving of every implementable policy on every gate, ~~and the paired tests are unequivocal~~ against the tuned threshold $p \leq 10^{-40}$ (better on 240/240 plan comparisons), ~~against the strongest rule-based trigger~~ $p \leq 10^{-40}$, ~~against the rollout policy~~

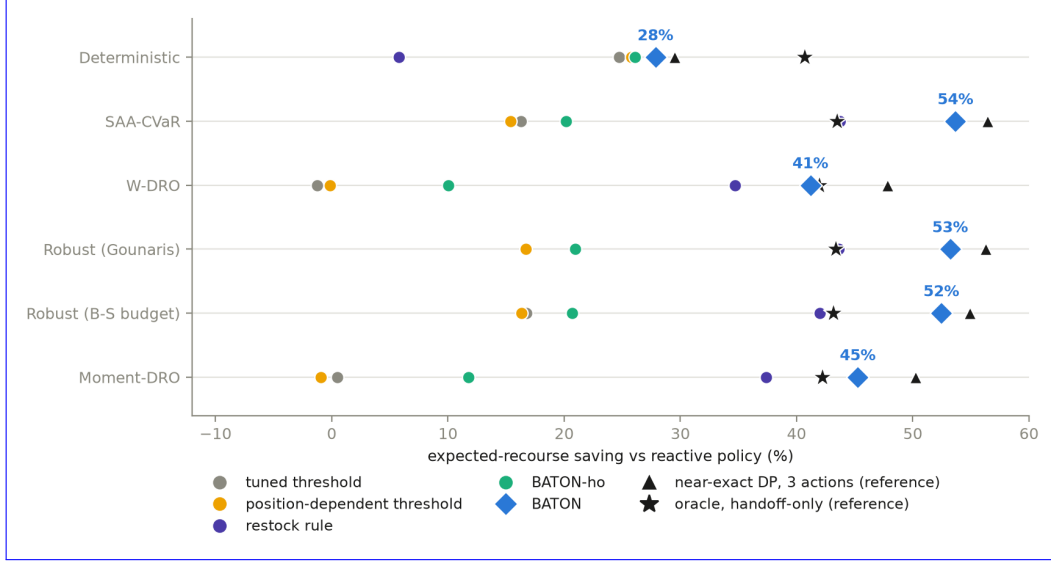


Figure 5: Expected-recourse saving by planning gate and execution policy (Table 2 drawn). BATON (diamonds) leads the implementable policies on every gate and exceeds the handoff-only clairvoyant bound (stars) wherever plans leave slack for the depot-return lever to work with.

~~$p \leq 10^{-40}$, against~~. Against the strongest competitor of each gate—the ~~position-dependent threshold on DET~~, depot-return restocking ~~$p \leq 10^{-38}$, against its own handoff-only restriction $p \leq 10^{-40}$, and against on the five conservative gates—its mean advantage ranges from 2.1 to 10.5 percentage points; the 95% confidence interval excludes zero on every gate, BATON is better on at least 31 of the 40 instances, and $p_{\text{Holm}} < 10^{-4}$ throughout. Against the near-exact ~~dynamic program—which three-action program, which sees fifty times its data but is confined to the handoff lever—~~ $p \leq 10^{-39}$ (better on 235/240), BATON attains 86%–96% of its saving; the shortfall is the finite-sample price of learning from a thousand days, and §4.7 shows that it closes as the history grows.~~

Second, the *composition* of the win changes across gates in an instructive way. On the deterministic gate, routes depart packed to capacity, most breach risk sits at the first stop where no online policy can act, and

all preventive policies compress toward each other; BATON’s margin there (~~27.8% against 24.8~~27.9% against 24.7% for the tuned threshold) comes mostly from the trigger correction of Proposition 2. On every conservative gate—the two sample-based gates and all three published robust planners—plans carry slack, breaches happen mid-route, and the depot-return lever becomes live: BATON ~~roughly doubles~~ more than doubles the saving of the best handoff-only policy (for instance ~~53.4% against 20.5~~53.7% against 20.1% on SAA plans) and ~~exceeds the handoff-only clairvoyant bound~~ exceeds the handoff-only clairvoyant bound (43.5%) on four of the six gates. ~~We emphasise what that comparison does and does not say: the bound is valid for the restricted action set, so~~ As explained in §3.6, that bound is clairvoyant only about the handoff lever: beating it is not a paradox but a ~~measurement—of exactly—measurement of~~ how much the second lever is worth when priced optimally.

Third, ~~richer threshold classes do not rescue~~ the threshold family ~~collapses where conservatism meets realistic prices~~. On WDRO ~~and M-DRO~~ plans the tuned threshold’s saving is ~~negative (-8.3~~negative or negligible (-1.3% and 0.5%): with handoffs carrying their true standby ~~vehicle-day cost, the thresholds’ interventions cost more than they prevent~~cost, while BATON ~~extracts 40.9% on the same plans~~its interventions cost about as much as they prevent. The position-dependent threshold, which can represent any monotone stopping boundary, does no better (-0.2% on WDRO); searching $m - 1$ boundaries directly against realised cost on a thousand days has more variance than estimating them by backward induction, and on DET plans, where it is strongest, it still trails BATON-HO (25.8% against 26.1%). The cost-scaled rollout, a threshold on the myopic cost with a tuned price multiplier, is the best handoff-only competitor on the conservative gates but ~~stays below BATON-HO everywhere~~. A policy that prices actions declines to act when acting has negative value; a policy that memorises a trigger level ~~cannot—~~, however flexibly, has to learn that boundary from realised cost

Table 4: Tail risk of the daily recourse bill of a plan (40 Dethloff instances per gate, 2,000 test days each). $CVaR_{95}$: mean of the worst 5% of daily plan bills; the reactive column gives its level in currency units, the others its reduction (%), higher is better). $P(\text{emg})$: share of days on which at least one route of the plan suffers an emergency (%), lower is better).

Gate	CVaR ₉₅ of daily plan bill				$P(\text{emg})$ per day (%)			
	reactive	thr.	BATON-HO	BATON	reactive	thr.	BATON-HO	BATON
DET	481.5	23.6	23.8	22.3	61.7	43.5	41.4	41.4
SAA	256.0	20.1	22.2	54.3	10.0	6.6	6.4	4.3
WDRO	58.9	1.3	7.2	37.9	2.3	1.6	1.7	1.3
ROB-G	280.2	21.3	23.2	52.2	11.3	7.7	7.6	5.2
ROB-BS	302.7	21.1	22.9	52.0	11.6	8.3	8.1	5.6
M-DRO	72.2	3.0	9.3	42.4	2.7	1.9	2.0	1.4

alone.

Fourth, the gains extend to the tail. Table 4 reports the mean of the worst 5% of daily plan-level bills and the probability that at least one route of a plan suffers an emergency on a given day. BATON cuts the $CVaR_{95}$ of the daily bill by 22.3%–54.3% relative to the reactive policy and, on the conservative gates, lowers the daily emergency probability more than any handoff-only policy, because a depot return prevents breaches that a priced handoff would not be worth. On DET plans the emergency probability of all policies stays high, again because the risk sits at the first stop.

The synthetic scenarios of Table 5 isolate the two defects of §3.3 under controlled conditions and confirm Proposition 1 at its extremal point: on collect-then-deliver structure the endpoint-trained policy saves exactly 0.0%—it never intervenes—while the label-corrected policies save about half of the reactive cost, ~~and~~ the option-value correction adds a further margin that grows with route length and the price ratio, and a depot return (priced flat at half a handoff) lifts BATON to 69.1%.

Table 5: Structural synthetic scenarios (five seeds, 1.2×10^4 test routes each, flat prices $\omega_F = 1$, C_{fail}/ω_F as shown or 5): saving % over the reactive policy, *higher is better*. The synthetic routes have no geometry, so the depot return of the full BATON is priced flat at half a handoff; the first three columns use the handoff lever only. On collect-then-deliver structure the endpoint-labelled predecessor never intervenes.

Scenario	endpoint + τ	peak + τ	BATON-HO	BATON
collect-then-deliver	0.0	49.2	51.9	69.1
milk run, regime switching	58.9	58.9	63.8	78.5
milk run, $C/\omega = 20$	85.5	85.5	88.2	93.5

4.3. Scale, real maps, and certified plans

Table 6 extends the comparison to the Salhi–Nagy family and the city instances. On Salhi–Nagy, whose 50–199-customer routes are tight and long, the picture of the conservative Dethloff gates repeats at scale: BATON saves 53.0% of reactive ~~cost—beyond cost—against 42.3% for its handoff-only restriction and beyond~~ the handoff-only oracle (~~49.0~~48.8%)—with the depot-return lever doing heavy work on the ~~centrally-located centrally located~~ depots of the CMT geometry.

On the city instances the same policy correctly reverses itself: ~~real~~. Real urban depots sit at the retail centre of gravity, but detours through congested networks are long, ~~and the delivery-dominated routes rarely come close to capacity late in the day, so~~ the return lever rarely pays, ~~deployment selection switches it off on most~~. Deployment selection keeps it in the menu on only 37% of the routes, and BATON equals its handoff-only restriction ~~within noise (10.8~~(10.6% against 10.6% across 19 instances) rather than paying the estimation cost of a useless action. ~~We regard this do-no-harm behaviour as a first-class experimental result: an operator can enable the full menu everywhere and trust the policy to discover, per route, which levers its geography supports~~The three-action policy’s value is therefore concentrated in settings with central depots and slack plans; on the dense urban networks we studied, the handoff lever does the work. Two reference points lie above BATON on the city instances, and neither is a competitor. The oracle knows

Table 6: Large-scale benchmarks under the fleet cost model (Det-gate plans; saving % vs. reactive, *higher is better*; the proposed policy is in bold; the last three columns are reference points). Salhi–Nagy instances carry 50–199 customers; the city instances (100–400 customers) place customers at real OSM shop locations on the road networks of Ho Chi Minh City, Hanoi, New York, Paris and Shanghai. The uniform twins use the same cities and demands with uniformly scattered customers; the deliver-only twins set the pickup of a random 25% or 50% of the customers to zero and are re-planned.

Benchmark	n	restock	thr.	thr.- k	BATON-HO	BATON	DP _{50k}	DP _{50k} ³	oracle
Salhi–Nagy	14	26.5	37.4	42.2	42.3	53.0	42.8	54.0	48.8
City, real shops	19	0.3	9.7	7.6	10.6	10.6	12.1	12.3	36.8
City, uniform	9	0.1	11.7	9.7	11.7	11.8	12.9	13.1	35.2
City, 25% deliver-only	19	-0.3	3.7	2.4	4.8	4.9	6.2	6.4	32.1
City, 50% deliver-only	19	-0.5	-1.5	3.3	3.1	3.1	6.8	6.7	38.3

the day’s demands in advance; on city routes much of the breach risk comes from demand surprises at the next stop that no policy observing only the past can anticipate, which is why its lead is large (36.8%). DP_{50k} is fitted on fifty times more data. With the return lever idle, the gap between it and BATON (12.1% against 10.6%) is the finite-sample cost of estimating the stopping boundary from 10³ days, and §4.7 shows it closing as the history grows. The uniform-scatter twins behave the same way.

The deliver-only twins test a feature of real demand that the benchmark protocol lacks. With a quarter or a half of the customers collecting nothing, the load rises less along the route, breaches become rarer, and every policy has less to save. BATON still saves 4.9% and 3.1%, while the tuned threshold falls to -1.5% at 50% deliver-only customers. In this setting, where recourse is worth least, the position-dependent threshold matches BATON within a few tenths of a point.

Figures 6 and 7 put these aggregates back on the map, using the same rendering of the real road network on which the instances were built; every trail in both figures follows the actual shortest road path between consecutive stops, so what is drawn is what the per-stop price schedules of §3.1 price. Figure 6 replays the demand-spike day of Figure 2 on the Hanoi network. The

planned vehicle (solid trail) works outward from the depot while its fitted continuation cost climbs with the accumulating pickups; at the circled stop the estimate crosses the handoff price and BATON calls the standby vehicle, which completes the dashed remainder of the route. The red cross marks where the reactive policy, facing the identical demands, instead runs the vehicle to overflow—mid-route, far from the depot, at the position where the emergency transfer is priced highest and the largest number of downstream customers is kept waiting. Figure 7 widens the lens from one route to an entire 200-customer plan on the same network: thirteen vehicles run simultaneously under BATON through one shared high-demand day. Seven routes hand off mid-route (circles; the standby vehicle continues on the dashed trail in the route’s colour), two absorb demand spikes too abrupt for any profitable anticipative action and fall through to the emergency guard, and the remaining four complete unaided. The figure makes the operating regime of the policy concrete: recourse is not an exceptional disaster response but a routine, priced re-balancing of work across the fleet, exercised on roughly half the routes of a hard day at a total recourse bill of a few hundred dollars.

The routing plans beneath every number above were certified against mixed-integer programming bounds, using the two-commodity flow formulation of Montané and Galvão (2006) solved by Gurobi with a 300-second budget per instance. The dual bounds certify the ALNS plans within 9.34% of optimality on average (worst case 12.55%), and at equal time budgets the solver’s own incumbents beat the heuristic plans on only 17/40 instances—so the execution-stage comparisons run on plans that are ~~simultaneously near-optimal~~close to optimal and representative of what strong planning practice produces. On 12–24-customer sub-instances, where the MIP proves optimality outright, even the fast construction heuristic used for the largest instances lands within 1.2–3.6% of the optimum.

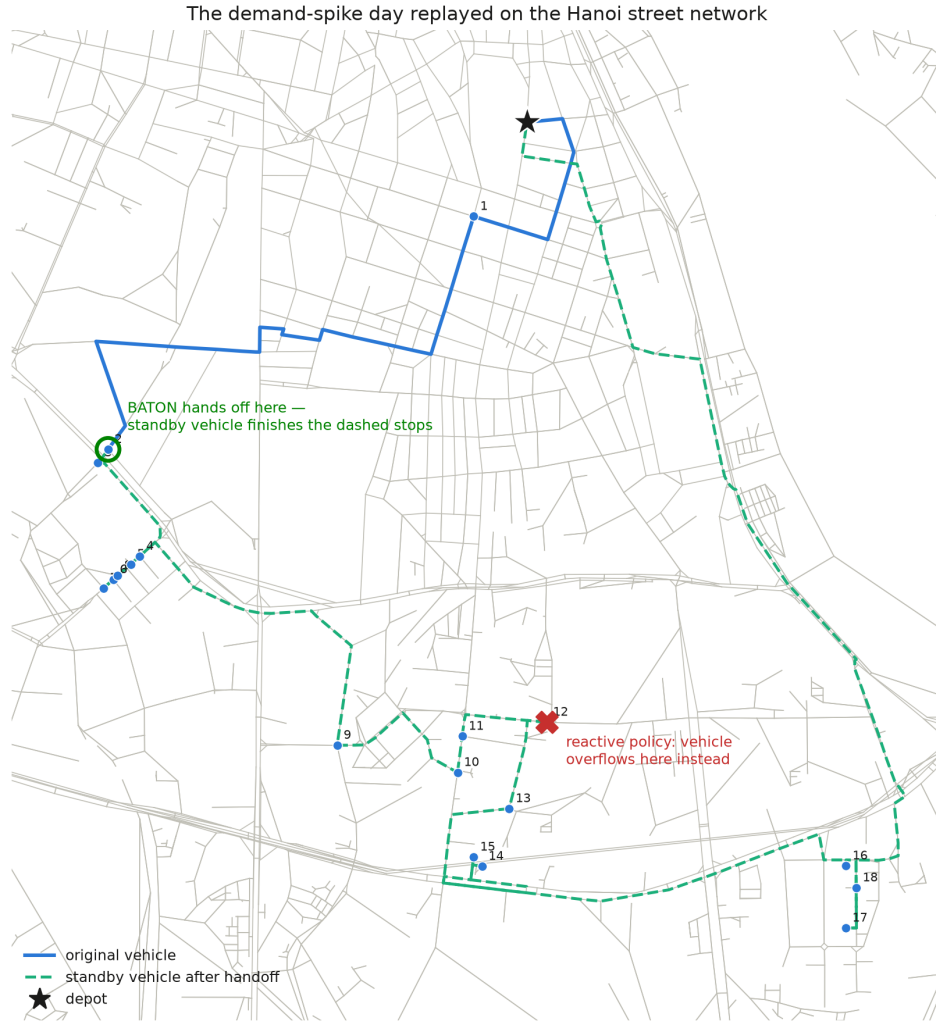


Figure 6: The demand-spike day of Figure 2 replayed on the Hanoi street network (instance HANOI-100-1; grey lines are the OSM drive network, the star is the depot, and every trail follows real shortest road paths). The planned vehicle drives the solid route; at the circled stop BATON’s estimated cost of continuing exceeds the handoff price and a standby vehicle takes over the dashed remainder. Under the reactive policy the same vehicle continues and overflows at the red cross, triggering the surge-priced emergency transfer with the rest of the route still unserved.

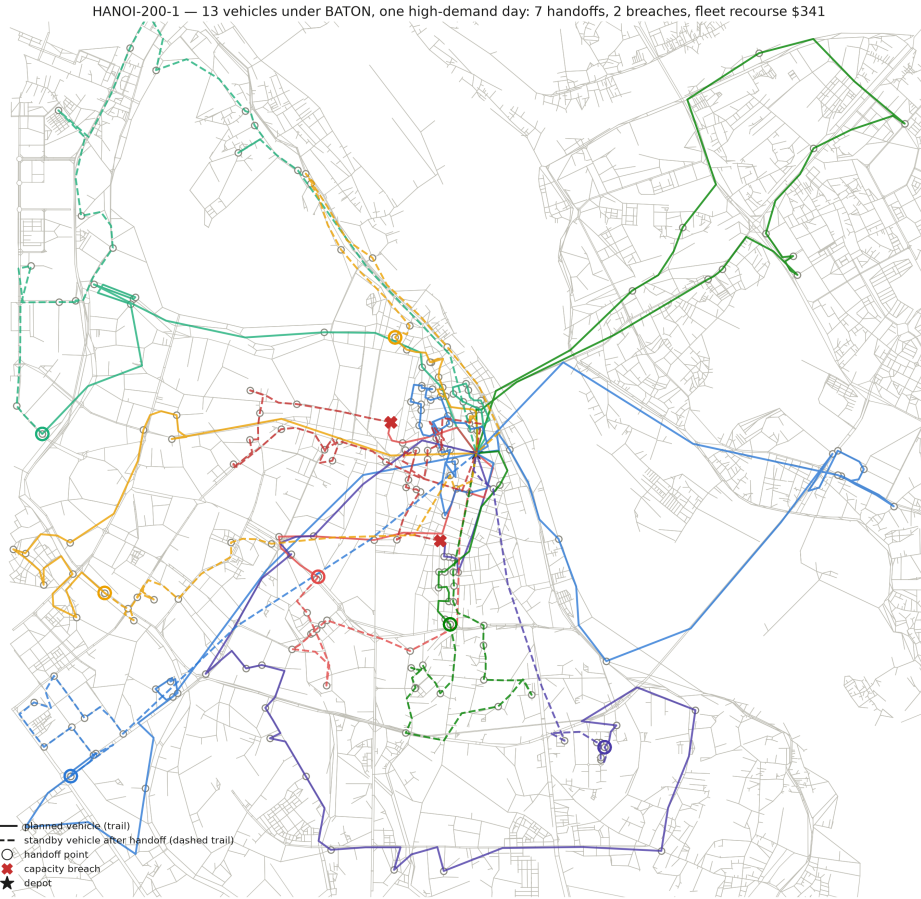


Figure 7: One high-demand day for a full 200-customer plan on the Hanoi network (instance HANOI-200-1): thirteen vehicles execute simultaneously under BATON, sharing one demand scenario. Solid trails are planned vehicles, dashed trails the standby vehicles that take over at the circled handoff points, and red crosses the two spikes that outran any profitable anticipative action and fell through to the emergency guard. Seven of thirteen routes exercised the handoff lever; the whole day's recourse bill is \$341.

4.4. *Against learning methods: ~~a GPU-trained policy~~, the price of over-triggering, and two deliberate negative results*

Because BATON is itself a learned policy, the natural challenge is a more flexible learner. We re-implemented the reinforcement-learning approach of Iklavov et al. (2024)—a policy network (two hidden layers of 64 units) over the observable state (position, load, slack, price ratios, remaining-demand moments), trained by REINFORCE with a ~~moving baseline—and trained it on a GPU over~~ moving-average baseline. Adam with learning rate 3×10^{-4} and batches of 256 days—over the identical routes and training scenarios available to BATON, evaluating both on identical held-out days. Table 7 reports the outcome for the specification’s 40 epochs and for 150 epochs under three seeds. The learned policy is respectable, clearly beating the rule-based family, but its best ~~configuration—after quadrupling the training epochs, adding entropy regularisation, and selecting over learning rates and seeds—saves 28.7~~ run saves 27.4% against BATON-HO’s ~~33.9~~ 32.9% on the same days, losing on ~~4039/50 routes ($p \leq 10^{-8}$)~~ 49 routes ($p \leq 10^{-7}$). The seeds cluster ~~within a percentage point of each other~~ closely and almost quadrupling the epochs barely moves the result, so the gap is not under-training; it is sample efficiency. A policy-gradient method must discover the value structure that the backward induction computes directly, and at a thousand paths per route the regression ~~wins—at five orders of magnitude less compute wins~~, at a small fraction of the compute (§4.7).

We have compared one learning architecture, and other architectures could narrow the gap. Value-based methods such as deep Q-learning estimate the same continuation value as BATON but with a generic function class; given the monotone structure of Proposition 4 and a scalar state, we would expect them to approach, not exceed, the isotonic regression at equal data. Actor-critic methods such as PPO reduce the variance of the policy gradient and would likely close part of the gap to BATON-HO by learning a critic that plays the role of $\hat{C}_k$. Attention- and graph-based architectures pay off when

Table 7: Reinforcement-learning baseline (re-implementation of the policy architecture of Iklassov et al., 2024) versus BATON on the identical 49 routes and out-of-sample test days; saving % vs. the reactive policy, *higher is better*. The learned policy acts on the handoff lever only, so BATON-HO is the like-for-like comparison; the full BATON row adds the depot return.

Policy	saving vs. reactive (%)
RL, 40 epochs	27.2
RL, 150 epochs, seed 0	27.4
RL, 150 epochs, seed 1	27.3
RL, 150 epochs, seed 2	27.3
BATON-HO (handoff only)	32.9
BATON (handoff and depot return)	33.7

the state is a large combinatorial object—which customers remain, where the standby vehicles are—and would become relevant for partial handoffs or fleet-level coordination, where the state is no longer a scalar. In the problem studied here the learner has little to discover that the backward induction does not compute exactly, and we therefore read the comparison as a statement about this regime rather than about reinforcement learning in general.

The price of over-triggering. Proposition 3 bounds what the myopic rule loses to the optimal one; Table 8 evaluates both sides on the Dethloff DET and SAA plans and the city plans, with both rules estimated on fifty thousand paths so that the comparison reflects the functional form rather than estimation noise. The pathwise ordering $\sigma^0 \leq \sigma^*$ holds on 100.0% of the test days, and the bound holds on 734/735 routes. Averaged over all routes the myopic rule’s regret is 3.8% of the reactive cost against a bound of 7.7%, and the clean-day term—the cost of handing off on days that would have ended without a breach—accounts for 90% of the bound, so unnecessary interventions are indeed what the myopic rule pays for. The breakdown answers the practical question of when a threshold is good enough. The regret is smallest

Table 8: The price of over-triggering (Proposition 3), per route, on 2,000 test days. The myopic rule and the optimal rule are both estimated on 5×10^4 paths (the myopic cost-to-go by the rollout regression, the optimal one by backward induction), so that the comparison reflects the functional form rather than estimation noise. Regret: excess expected cost of the myopic rule; bound: right-hand side of (13); clean-day term: $\mathbb{E}[H_{\sigma^0} \mathbf{1}\{\sigma^0 < \sigma^*, T = \infty\}]$; all in % of the reactive cost. $P(\sigma^0 < \sigma^*)$: share of days on which the myopic rule hands off before the optimal one; containment: share of days with $\sigma^0 \leq \sigma^*$. Last column: excess cost of the tuned threshold over BATON-HO, both fitted on 10^3 days (% of reactive cost).

Routes	n	$\bar{m}$	regret	bound	clean-day term	$P(\sigma^0 < \sigma^*)$	$\sigma^0 < \sigma^*,$ $T = \infty$	contain.	thr. – BATON-HO
Dethloff, DET	256	7.8	3.8	7.7	6.9	16.3	4.7	100.0	1.0
Dethloff, SAA	320	5.9	6.8	10.9	8.4	1.6	0.4	100.0	3.0
City, DET	159	17.0	2.0	4.8	5.7	1.7	0.8	100.0	0.4
all, $m \leq 12$	536	5.8	2.7	5.9	5.7	5.6	1.6	100.0	0.8
all, $13 \leq m \leq 18$	132	15.6	6.8	12.3	9.9	12.2	3.5	100.0	1.7
all, $m \geq 19$	67	20.5	4.5	9.2	8.9	4.9	2.0	100.0	0.9

on tightly packed deterministic plans (3.8% of reactive cost) and on short routes of at most 12 stops (2.7%), and larger on the conservative SAA plans (6.8%) and on longer routes (6.8% and 4.5% for 13–18 and 19 or more stops). The same pattern holds for the tuned threshold fitted on a thousand days: its excess cost over BATON-HO is 1.0% of reactive cost on DET plans, 0.4% on the city plans and 3.0% on SAA plans. On tight plans and on the city routes a tuned threshold is thus a defensible simplification of the handoff decision; on plans that carry slack, the functional form costs money—and the larger loss there comes from the missing depot return, which no handoff threshold can price.

Two negative results. Two further experiments were designed to fail, and did, in ways that sharpen the method’s justification. First, we enriched the conditioning statistic beyond the scalar W_k with exactly the features a representation-learning critique would demand: the precision-weighted residual that is the posterior location of the day’s common demand factor, the last increment, and the running maximum increment, fitted with monotonicity-constrained gradient-boosted trees. On 25 real routes the enriched policy *lost*

to the scalar isotonic policy on 23, by 3.2% of cost on average, while fitting orders of magnitude slower: at operational sample sizes the extra features buy more estimation variance than signal, and Proposition 4’s shape constraint is doing the statistical heavy lifting. Second, the equal-data plug-in dynamic program—the same backward induction with quantile bins in place of the isotonic projection—collapses exactly where data are scarcest relative to risk (on ~~conservative~~ [WDRO](#) and [M-DRO](#) plans it saves ~~nothing or worse~~; [Table 20.0%](#) and [0.0%](#), against [10.0%](#) and [11.8%](#) for [BATON-HO](#)), because bins cannot pool information across the load axis the way the monotone projection can. Together the two null results locate BATON at a complexity sweet spot: one scalar statistic, one shape constraint, no free parameters.

4.5. [Robustness to the modelling assumptions](#)

[This subsection probes three assumptions behind the model: that the scalar load is a sufficient state, that days are exchangeable, and the treatment of the post-return state.](#)

[Dependent demand and the sufficiency of the scalar load.](#) [Table 9 holds the Dethloff DET and SAA plans fixed and re-fits and re-tests every policy under five demand laws: Gaussian-copula correlations \$\rho = 0, 0.3, 0.6, 0.9\$ among the deliveries and among the pickups, and a route-level multiplicative day factor on top of independent marginals. Three findings stand out. First, dependence helps rather than hurts. Every policy saves more as \$\rho\$ grows, because correlated demand creates heavy days whose breaches can be foreseen from the running load; under independence much of the risk is idiosyncratic and cannot be anticipated. Second, BATON keeps the highest saving among the implementable policies under every law, with one tie within noise discussed below. On DET plans it attains 100%, 94% and 93% of the near-exact three-action saving at \$\rho = 0, 0.6\$ and \$0.9\$. Third, the monotone shape survives strong dependence. On fifty thousand paths per route, the unconstrained binned estimate of the conditional expected future cost given \$W_k\$ decreases](#)

Table 9: Robustness to the demand dependence (40 Dethloff instances, DET and SAA plans, saving % vs. reactive; plans are held fixed and policies are re-fitted and re-tested under each demand law; react.: expected daily recourse cost of the reactive policy per plan, which shows how much there is to save). ρ is the Gaussian-copula equicorrelation among deliveries and among pickups; the day-factor law multiplies every demand of a day by a common lognormal factor (s.d. 0.25) on top of independent marginals. Last column: share of adjacent-bin pairs in which the unconstrained binned estimate of $\mathbb{E}[\text{future cost} \mid W_k]$, fitted on 5×10^4 paths per route with 25 bins, *decreases* significantly (one-sided, 2.5% level; the rate under independence, where monotonicity is a theorem, calibrates the test).

Demand law	gate	react.	thr.	thr.- k	BATON-HO	BATON	BATON-CF	DP _{50k} ³	oracle	viol. (%)
$\rho = 0$ (independent)	DET	176.4	9.9	10.5	10.6	15.5	15.4	15.5	40.1	0.2
	SAA	2.8	-24.9	0.0	-8.0	-0.1	0.2	7.4	37.3	
$\rho = 0.3$	DET	180.2	18.0	19.0	19.3	21.9	22.5	22.7	40.6	–
	SAA	8.0	3.1	1.3	7.9	36.3	36.6	40.9	42.9	
$\rho = 0.6$ (benchmark)	DET	179.2	24.7	25.8	26.1	27.9	29.2	29.5	40.7	0.1
	SAA	16.5	16.3	15.4	20.1	53.7	53.8	56.4	43.5	
$\rho = 0.9$	DET	177.5	32.3	33.9	34.1	35.4	37.6	38.0	40.7	0.0
	SAA	25.9	27.3	29.1	33.0	64.5	65.9	67.7	43.8	
day factor, $\rho = 0$	DET	172.2	9.1	9.7	9.8	14.2	14.2	14.2	40.0	–
	SAA	5.2	-15.6	-0.1	-6.9	7.4	8.0	12.4	41.7	

significantly between adjacent bins in 0.2% of pairs under independence, where Proposition 4 guarantees monotonicity, and in 0.1% and 0.0% of pairs at $\rho = 0.6$ and 0.9. The isotonic restriction is therefore justified empirically where the theorem does not strictly apply.

The exception is instructive. Under independence the SAA plans, calibrated for correlated demand, are so conservative that the reactive policy costs only 2.8 per plan-day and breaches are rare in a thousand training days. Out of sample, the handoff interventions of the tuned threshold and of BATON-HO then cost more than they prevent (-24.9% and -8.0%). The position-dependent threshold learns never to act (0.0%), as does BATON (-0.1%), and BATON-CF is marginally positive (0.2%). This is the small-data regime of Remark 1: with a few dozen breaches per history, backward induction propagates noise, and a policy that declines to act is the right outcome.

Non-exchangeable days. The model treats days as exchangeable, whereas real demand has seasons, weekdays and promotions. Table 10 makes one day in five a promotion day, with mean deliveries 15% and mean pickups 35%

Table 10: Non-exchangeable days (40 Dethloff instances, DET and SAA plans; saving % vs. reactive on the same days). One day in five is a promotion day with mean deliveries $\times 1.15$ and mean pickups $\times 1.35$. Pooled: one fit on a history that mixes both day types; day-type-specific: separate fits, the day type being known in advance (promotions are scheduled); stale: fitted on normal days only and applied unchanged on promotion days.

Fitting strategy	tuned threshold			BATON		
	normal	promotion	all	normal	promotion	all
pooled fit (day type ignored)	21.2	22.7	21.7	26.8	29.4	27.6
day-type-specific fits	21.3	23.2	21.9	26.8	30.9	28.1
stale fit (normal days only)	21.3	22.2	21.6	26.8	28.4	27.3

above normal—returns surge after promotions—and compares three ways of fitting the policy. A pooled fit ignores the day type; day-type-specific fits use separate histories, as an operator can when promotions are scheduled; a stale fit is trained on normal days only and applied unchanged on promotion days. On promotion days BATON saves 30.9% with a day-type-specific fit, 29.4% with the pooled fit and 28.4% with the stale fit; on normal days the three coincide. The policy thus degrades gracefully under a shift it was not trained for, because its trigger compares costs at the current load rather than a memorised level, and a separate fit per known day type recovers the remainder at a cost of milliseconds. The tuned threshold gains little from any strategy, because its single level cannot track the shift either way.

The fresh-start value and the post-return state. Table 11 examines the depot-return machinery of §3.5 on the three settings where it matters most or least: Salhi–Nagy, where returns are used on about a fifth of the days; Dethloff SAA plans; and the city routes, where they rarely pay. To expose the three-action fit, all columns are computed *without* deployment selection. The first two columns measure the error of BATON’s fresh-start value against the near-exact value from fifty thousand independent paths. In-sample, the error is +0.50% of the handoff price on Salhi–Nagy and +0.00% on Dethloff; out of sample, for the same fitted downstream policy, it is +0.09% and +0.01%. Biases (i) and (ii) of §3.5 are therefore small and nearly cancel,

Table 11: The fresh-start value and the post-return state. Bias columns: mean error of the estimated fresh-start value $\hat{F}_k$ against the near-exact F_k of DP_{50k}^3 , in % of the handoff price; in-sample is BATON’s own estimate, out-of-sample re-evaluates the same fitted downstream policy on 5×10^4 independent paths. Saving columns (% vs. reactive, all *without* deployment selection, so that the three-action fit is exposed): the three-action policy as fitted, with $\hat{F}_k$ replaced by its out-of-sample value, with F_k replaced by the near-exact value, with the state-conditional fresh-start value (BATON-CF), and with the exact residual-capacity reset $x_k = -D_{\leq k}$.

Benchmark	bias of $\widehat{F}_k$ (% of H)		three-action BATON, saving %						
	in-sample	out-of-sample	BATON-HO	as fitted	F out-of-s.	F near-exact	CF	exact reset	DP $^3_{50k}$
Salhi–Nagy (Det)	+0.50	+0.09	41.3	52.2	52.2	51.9	53.1	57.9	53.5
Dethloff (SAA)	+0.00	+0.01	18.9	51.4	51.4	51.4	52.2	54.6	54.6
City, real shops (Det)	-1.18	-1.05	9.2	5.6	5.6	5.9	9.4	9.9	10.7

and replacing $\hat{F}_k$ by its near-exact value changes the saving by a fraction of a point (52.2% against 51.9% on Salhi–Nagy). Bias (iii) is different. On the delivery-dominated city routes the three-action fit without selection saves only 5.6%, below the handoff-only policy (9.2%): the running load drifts downward there, a return is chosen only on unusually heavy days, and an unconditional fresh-start value under-prices it on exactly those days. The state-conditional fresh-start value removes the effect (9.4%) and slightly improves the other two settings as well, so BATON-CF performs as well as BATON *with* deployment selection without needing it. The last policy column prices the exact post-return state $x_k = -D_{\leq k}$ instead of the conservative reset to zero. It raises the saving to 57.9% on Salhi–Nagy and 54.6% on Dethloff: the conservative convention of the main tables understates what the depot return is worth, so the three-action results we report are conservative. (The exact reset can exceed DP_{50k}^3 , which is computed under the conservative convention and is therefore a reference for that convention only.)

4.6. *Shared, capacitated standby capacity*

The default model treats standby capacity as pay-per-use from a pool large relative to one plan (§3.1): each route then decides alone. When instead a fixed number S of standby vehicles is reserved for the day, their holding cost F_{sb} is paid whether or not they are used, a handoff costs only its

marginal price $H_k - F_{\text{sb}}$, and the routes of a plan compete for the pool. We simulate this coupling exactly. On every test day the routes of a plan run simultaneously, handoff requests are served first come, first served in the order of the time at which they arise along the routes, and a route whose request is refused continues under its fallback menu (continue or return to the depot, fitted by the same backward induction with the handoff removed). Two policies are compared. The *naive* policy decides every route at the marginal price and ignores the cap. The *shadow-priced* policy decides every route at the price $H_k - F_{\text{sb}} + \lambda$, where the shadow price $\lambda \geq 0$ of a standby vehicle is chosen, for each plan and pool size, on the training days; it decouples the routes, which can then still be decided one at a time. For each plan we compute the expected daily total—holding plus recourse—for every pool size from $S = 0$ to one vehicle per route, and report the cost-minimising size S^* .

Table 12 gives two answers. The first concerns the pool size. At the default holding cost, reserving even one standby vehicle for a single plan never pays: the cost-minimising pool size is $S^* = 0$ for every plan, because the expected number of handoffs per plan-day is far below one, and a vehicle held for one plan mostly waits. (Under pay-per-use, the pool that covers 95% of days has a median of 0 and a maximum of 7 vehicles per plan for BATON, and 1 and 8 for BATON-HO, whose only lever is the handoff.) Standby capacity is economical only when it is shared across many plans, so that its holding cost is spread over the handoffs of the whole operation, which is the pay-per-use regime of our default model. The second answer concerns the coupling. Given a cap, deciding each route with a shadow price is clearly better than ignoring the cap. A naive route requests handoffs it may be refused, and a refused route has already let its load climb past the point at which a return would have paid. With no standby vehicle at all, shadow pricing lowers the daily recourse cost by 15.8% on the Dethloff plans and 4.1% on the city plans; with one vehicle for the plan, by 9.1% and 3.5%. The

Table 12: Shared, capacitated standby pool: expected daily recourse cost per plan (currency units, excluding the holding cost of reserved vehicles). Pay-per-use: the default model (pooled capacity, day rate billed per handoff). Reserved pool of S vehicles: each held vehicle costs $F_{\text{sb}} = 20$ per day whether or not it is used, a handoff then costs its marginal price $H_k - F_{\text{sb}}$, requests are served first come, first served in the order of their time along the routes, and a route whose request is refused continues under its fallback menu (continue or return to the depot). Naive: every route decides at the marginal price and ignores the cap. Shadow: every route decides at $H_k - F_{\text{sb}} + \lambda$, with the shadow price λ of a standby vehicle chosen on the training days. S^* : pool size minimising holding plus recourse for the shadow-priced policy.

Benchmark	plans	routes	pay-per-use	$S = 0$		$S = 1$		$S = 2$		S^*
				naive	shadow	naive	shadow	naive	shadow	
Dethloff (SAA plans)	40	9.5	8.3	9.9	8.3	9.0	8.2	8.5	8.1	0.0
City (Det plans)	19	8.4	33.4	38.3	36.7	34.2	33.0	31.4	30.5	0.0

shadow price decouples the routes, so the policy is still fitted one route at a time, and the depot return is the lever that makes a small or empty pool tolerable.

4.7. *Data and computing budgets*

Every data-driven policy in the preceding tables was fitted on the same 10^3 training days per instance; only the two reference programs saw more. Table 13 and Figure 8 vary the history from 10^2 to 2×10^4 days per route, on the same test days, for BATON, the label-corrected tuned threshold and the equal-data dynamic program.

BATON beats the tuned threshold at every history length we tested. With only 100 days per route it saves 25.8% on DET plans and 41.7% on SAA plans, where the threshold saves -2.4% and the equal-data dynamic program is no better. The isotonic projection pools information along the load axis, whereas quantile bins need many paths per bin before their estimates stabilise, which is why DP_N trails even the threshold on short histories. As the history grows, BATON approaches the near-exact references. On the SAA plans it reaches 56.3% at 2×10^4 days against 56.4% for DP_{50k}^3 , and on the city routes it slightly exceeds the binned references (12.9% against 12.2%), the

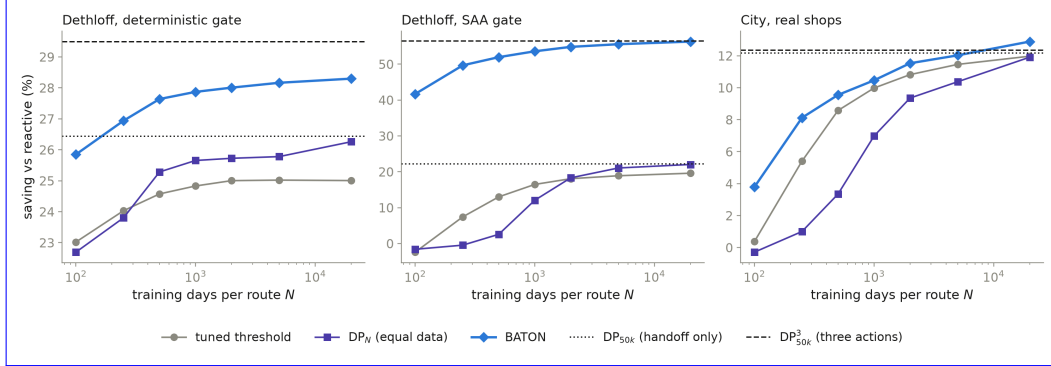


Figure 8: Saving over the reactive policy as a function of the number of training days per route (log scale), on identical test days. Horizontal lines: the near-exact handoff-only and three-action dynamic programs fitted on 5×10^4 days.

binning programs being only near-exact themselves. The gaps in Table 2 between BATON and the references are therefore the finite-sample cost of learning from a thousand days, not a structural shortfall of the policy. The clairvoyant oracle’s lead, by contrast, does not shrink with data: it is the value of knowing the future.

Table 14 separates offline from online computation on one CPU core. Offline, fitting BATON-HO takes 7 ms per route and the full policy, including the fresh-start valuations and deployment selection, 27 ms; the grid-tuned competitors cost about the same or more, because tuning re-simulates the training days for every candidate parameter. Online, every cost-comparison policy makes one lookup in a monotone step function and two comparisons per stop, about $1 \mu\text{s}$ per decision through the fitted breakpoints. The reinforcement-learning baseline of §4.4 is the exception: on the same CPU its training takes 1–3 minutes for the 49 routes of Table 7, against 1.7 seconds for fitting BATON on the same routes, so the comparison of Table 7 is, if anything, generous to the learner in computing budget and identical in data.

4.8. Economic and spatial sensitivity

A recourse policy earns its keep only if its advantage survives changes in the prices that define it. Table 15 and Figure 9 sweep the four manage-

Table 13: Training-data budget: saving % vs. reactive as a function of the number N of training days per route (same test days throughout). thr.: label-corrected tuned threshold; DP_N : plug-in dynamic program at equal data; BATON with deployment selection. The last row gives the near-exact references (handoff-only / three-action).

N	Dethloff, DET			Dethloff, SAA			City, DET		
	thr.	DP_N	BATON	thr.	DP_N	BATON	thr.	DP_N	BATON
100	23.0	22.7	25.8	-2.4	-1.6	41.7	0.4	-0.3	3.8
250	24.0	23.8	26.9	7.4	-0.5	49.7	5.4	1.0	8.1
500	24.6	25.3	27.6	13.0	2.6	52.0	8.6	3.3	9.6
1,000	24.8	25.7	27.9	16.5	12.1	53.6	10.0	7.0	10.5
2,000	25.0	25.7	28.0	18.1	18.3	54.8	10.8	9.4	11.5
5,000	25.0	25.8	28.2	18.9	21.1	55.6	11.5	10.4	12.0
20,000	25.0	26.3	28.3	19.6	22.0	56.3	12.0	11.9	12.9
ref.: DP_{50k} / DP_{50k}^3		26.4	29.5		22.2	56.4		12.2	12.3

Table 14: Offline fitting time per route (single CPU core, milliseconds; median and 90th percentile over 102 routes of ten Dethloff SAA plans and four city plans). Every policy is fitted on the same 10^3 training days except the two reference programs. Tuning by grid search on realised training cost is included. The online decision of every cost-comparison policy is one lookup in a monotone step function and two comparisons: $1.2 \mu s$ per decision through the fitted breakpoints ($66 \mu s$ through the scikit-learn prediction call).

Policy	training days	median	p90
tuned threshold (peak label)	10^3	62.8	105.1
position-dependent threshold	10^3	67.9	171.3
π_3 rule (grid-tuned)	10^3	5.1	8.0
rollout	10^3	7.3	10.8
cost-scaled rollout	10^3	50.9	78.9
depot-return restocking	10^3	5.1	7.5
DP_N (equal data)	10^3	3.4	5.3
BATON-HO	10^3	7.5	12.5
BATON (incl. deployment selection)	10^3	27.5	52.4
BATON-CF	10^3	29.3	60.2
DP_{50k} (reference, 5×10^4 paths)	5×10^4	113.2	181.2
DP_{50k}^3 (reference, 5×10^4 paths)	5×10^4	299.3	626.9

rial cost levers one at a time around the defaults: the standby day rate, the emergency callout, the surge multiplier, and the per-customer lateness price. BATON leads in every configuration, but the instructive content is

Table 15: Sensitivity of the recourse saving (% , *higher is better*; best non-clairvoyant value per row in bold) to the fleet-economics parameters, one factor at a time around the defaults (12 Dethloff instances, DET and SAA gates; the baseline row is computed on the same 12 instances). BATON re-balances its action mix as prices move and leads in every configuration. In the last row the standby rate exceeds the emergency price at late stops, so Inequality (7) fails there.

Configuration	restock	threshold	BATON-HO	BATON	oracle (HO)
baseline	28.2	24.9	29.0	48.0	44.5
cheap emergencies ($F_{\text{emg}}=25$)	23.4	13.0	18.0	40.1	31.0
dear emergencies ($F_{\text{emg}}=60$)	32.9	36.0	39.6	55.2	55.3
cheap standby ($F_{\text{sb}}=10$)	28.2	37.5	41.7	52.7	57.7
dear standby ($F_{\text{sb}}=35$)	28.2	8.8	13.1	42.6	24.8
low SLA price ($p_{\text{late}}=0.5$)	33.1	22.3	26.4	49.0	42.0
high SLA price ($p_{\text{late}}=3$)	23.6	28.5	32.3	47.0	47.6
mild surge ($s_{\text{emg}}=1.5$)	27.4	20.6	24.8	44.7	40.3
heavy surge ($s_{\text{emg}}=4$)	29.9	30.4	34.1	52.0	49.5
standby above emergency ($F_{\text{sb}}=60$)	28.2	1.4	2.1	36.8	5.0

how. When standby capacity is dear ($F_{\text{sb}} = 35$), the threshold family collapses to 7.38.8%—its handoffs cost more than they save—while BATON holds 41.942.6% by shifting its mix toward depot returns, ending far above the handoff-only clairvoyant bound (24.8%). When lateness is expensive, depot returns lose their appeal and the mix shifts back toward handoffs. The last row sets the standby rate so high ($F_{\text{sb}} = 60$) that a handoff costs more than an emergency at late stops, violating Inequality (7). Nothing in the theory depends on that inequality (Assumption 2), and the policy behaves accordingly: it stops handing off where the handoff is dominated, relies on depot returns, and saves 36.8%, while the tuned threshold saves 1.4% and the handoff-only restriction 2.1%. No parameter changes across these worlds; the policy simply re-prices its levers.

The spatial-structure experiment closes the study. For each city instance we generated a twin with identical demands and identical seeds but customers scattered uniformly over the street network rather than placed at real shops; Figure 10 shows the pair for the 200-customer Hanoi instance, and the visual difference is the experiment: real retail concentrates around the commercial

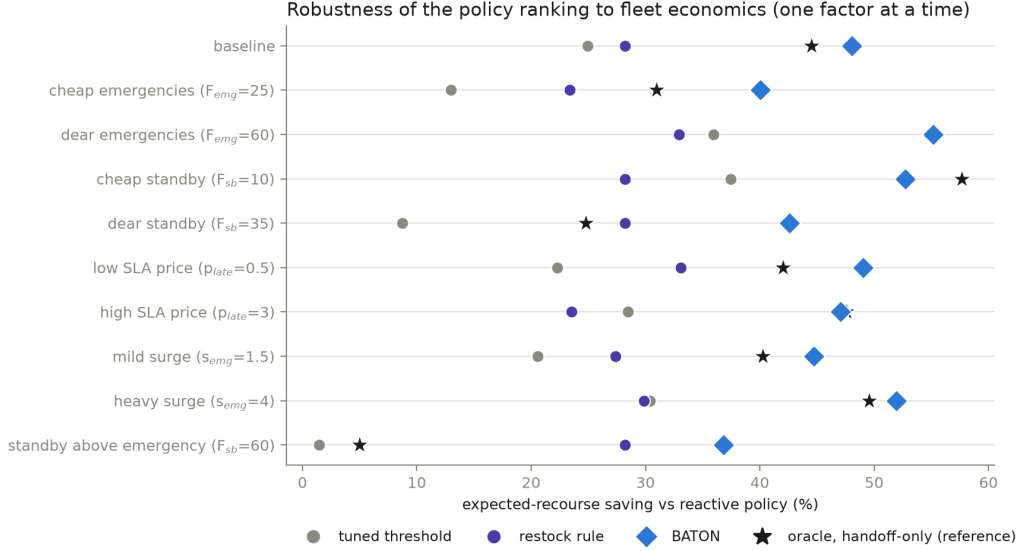


Figure 9: Sensitivity of the policy ranking to the fleet economics, one factor at a time. BATON (diamonds) leads in every configuration, re-balancing between handoffs and depot returns as the relative prices move.

core while the uniform convention spreads demand into residential and peripheral blocks. The real retail clustering shortens optimal routes by 25.9% on average—uniform spatial conventions materially overstate travel in urban simultaneous pickup-and-delivery—while the execution-policy ranking and BATON’s margins are unchanged between the twins. The policy conclusions of this paper do not depend on where customers stand; the absolute economics do, which is precisely why we report both.

5. Managerial insights and conclusions

This paper set out from an operational nuisance—mid-route capacity breaches in simultaneous pickup-and-delivery—and arrived at a **general** prescription for execution-stage recourse: *price the levers, estimate the cost of waiting, and compare*. The resulting policy, BATON, is a data-driven optimal-stopping rule that fits in milliseconds per route from ordinary historical de-



Figure 10: The spatial-structure twins for HANOI-200-1 on the OSM drive network (star: depot). Left: the 200 customers at real OpenStreetMap shop locations, clustered along the retail corridors around the commercial centre. Right: the uniform twin—identical demand draws, customers scattered uniformly over the same street network. Real clustering shortens optimal routes by 25.9% on average while leaving the execution-policy ranking unchanged.

mand logs, carries no tuning parameter, and accommodates ~~whatever the~~ recourse menu an operation actually offers. Across six planning regimes, three benchmark families including instances built on real road networks and real retail locations, and ~~eleven~~ competitors spanning the rule-based, threshold, restocking, rollout, dynamic-programming, and reinforcement-learning traditions, it delivered the lowest ~~execution cost everywhere, approached expected execution cost of every implementable policy, up to a tie in the setting where recourse is worth least, and it reached 86%–96% of the saving of~~ a near-exact dynamic program given fifty times its data, ~~and~~. Where plans leave room for the depot return it also exceeded the clairvoyant bound of the ~~single-lever policy class wherever plans left room for its second lever to work~~ handoff-only class; that bound is clairvoyant only about the handoff lever, so the excess is a measure of what the second lever is worth, not a contradiction.

For the practitioner, ~~four~~ five findings travel beyond our test bed. First, *label on the peak*. Aggregated per-route data invite risk labels computed from end-of-route totals, and Proposition 1 shows the resulting policies can be blind to the very event they are meant to prevent; monitoring must record the running maximum. Second, *thresholds over-trigger by construction*. ~~The one-sided error of Proposition 2 means every tuned trigger level buys its simplicity with premature interventions, and the cost of that simplicity explodes precisely when intervention prices are high—in our sweeps~~ the premium can be predicted. Proposition 3 bounds it by the handoff price times the probability of interventions on days that would have completed cleanly. On tightly packed plans and short routes that premium is small, and a tuned threshold is a defensible simplification of the handoff decision; on plans that carry slack it is larger, and there the threshold family destroyed value ~~on conservative plans in our experiments~~ while the priced policy still earned double-digit savings. Third, *planning conservatism and execution intelligence are substitutes, but asymmetric ones*: robust plans leave ~~little~~ less recourse cost to save, yet what remains is recovered almost exclusively by the policy that prices the full action menu. ~~An operator choosing between a more conservative planner and a smarter execution layer should note that the latter kept earning when the former had nothing left to give.~~ Fourth, *geography selects the lever*. Depot returns dominate where depots are central—classical benchmark geometries flatter ~~them—and them—and~~ are nearly worthless on real urban networks ~~with peripheral depots~~; a deployed policy must discover this per route, and on our city instances BATON ~~'s train-set deployment selection does so at no risk to performance where the extra lever is useless~~ does so and reduces to its handoff-only form. The three-action policy's value is therefore concentrated in settings with central depots and slack in the plans, not in the dense urban networks we studied. Fifth, *pool standby capacity*. At our prices a standby vehicle reserved for a single plan never pays for its holding cost, because a plan needs far less than one handoff per day; standby

capacity is economical when it is shared across many plans. Where capacity is nonetheless capped, deciding each route with a shadow price on the pool beats ignoring the cap, keeps the routes decoupled, and relies on the depot return as the fallback lever (§4.6).

~~The study has limits that chart the next steps.~~

Scope of the conclusions. Our managerial conclusions hold for the model we studied, and several features of real operations lie outside it. The model has no time windows: lateness is priced per downstream customer, uniformly, and customers with heterogeneous delay penalties or hard windows would make the handoff and return prices depend on which customers remain, not only on how many. The routes are fixed and executed in their planned order; we do not re-sequence the remaining stops, and we do not integrate execution with route planning. Shared resources enter only through the standby pool of §4.6, with a first-come-first-served dispatch rule and a shadow price chosen per plan; depot congestion, driver hours and standby vehicles that serve several routes in one day are not modelled. Demand is simulated from a calibrated protocol rather than drawn from operational logs, and the fleet costs are calibrated to orders of magnitude rather than to one operator’s rate card. The comparisons between execution policies are, we believe, robust to these choices, since every policy faces the same model and our sensitivity analyses move each of them; the absolute savings are not, and should be read as indicative.

Limitations and next steps. The policy conditions on a scalar load statistic; ~~our~~. Our negative result shows ~~that~~ richer statistics do not pay at a thousand training paths, and the robustness study shows that the scalar state loses little under strongly dependent demand, but fleets with vastly longer logs may cross that boundary, and the recursion accepts any regressor. The action set ~~is~~ while ~~is~~ general in form ~~is~~ was instantiated with two levers; ~~partial handoffs—transferring~~ but has definite boundaries. Proposition 4 and the

algorithm cover any action that ends the route at a price known at the current stop or resets the load to a known level at a known price; handoffs and depot returns are of these kinds. A *partial* handoff—transferring only the riskiest remaining customers—compose naturally with the same backward induction and await customers to a standby vehicle—is not: it changes the remaining route itself, so the state must record which suffix of the route the planned vehicle still serves, the price schedule must be recomputed for every possible split, and the continuation value becomes a family of functions indexed by the split point. The backward induction extends to that augmented state with one regression per stop and split, at a cost quadratic rather than linear in the route length, and awaits a per-customer cost model to be worth instantiating. The demand model treats days as exchangeable; systematic seasonality would enter through day-type-specific fits. The experiment of §4.5 shows what systematic variation does to a deployed policy: a policy fitted on ordinary days and applied unchanged on promotion days keeps most, but not all, of its saving, while fitting by day type recovers the rest. Where the day type is known in advance—promotions, seasons, weekdays—the remedy is therefore to fit one policy per day type, which costs milliseconds; where it is not, the policy should be re-fitted on a rolling window and monitored for drift. Finally, our deepest structural finding cuts toward planning: on deterministically packed routes most breach risk sits at the first stop, where no execution policy can act, so the larger prize is a planner whose acceptance gate internalises the fitted execution cost itself—co-optimising the routes with the recourse policy that will execute them. The machinery developed here, cheap enough to sit inside a We regard this as the most important direction for future work, because it attacks the risk that no execution policy can reach, and we did not attempt it here for two reasons. It changes the object of study—the plans would then depend on the execution policy, so a comparison of execution policies on common plans, which is the design that isolates the contribution of this paper, would no longer be possible—and it requires the fitted execution

cost of every candidate route inside the planner’s inner loop, makes which is a separate computational question. The machinery developed here is cheap enough to make that co-optimisation a concrete ~~and, we believe, imminent~~ possibility.

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Data availability

All instance generators, result files, and code sufficient to reproduce every table and figure are available in a public repository at https://github.com/vinhqddang/stochastic_vrp. Machine-checked proofs of Propositions 1–4, written in Lean 4 with the Mathlib library, are in the same repository, in the subdirectory BatonProofs of the manuscript directory.

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